

Vehicle Financing System (Origination)

Functional Specification Document (FSD)

Revision History

Date	Version	Author	Description
21-05-2026	1.0	Rajkumar Singh Sengar	v.1. (Blox as sole Financer)
16-07-2026	2.1	Meenakshi	V.1.1. (Incorporated Multi-financer flow & Sales assisted Journey)

1. Executive Overview & Purpose

This document defines the functional specifications for the Vehicle Financing System (Origination) module, a core component of BLOX's Digital Shariah-Compliant Vehicle Financing Platform. It governs the complete lifecycle of vehicle financing—from customer onboarding and application submission through KYC verification, credit assessment, offer generation, agreement execution, disbursement, LMS booking, and vehicle delivery—within a Qatar-hosted cloud environment operating under Qatar Central Bank (QCB) regulatory requirements.

The solution is structured using a microservice-based architecture and a centralized Business Rules Engine (BRE) framework, providing clear functional ownership, process governance, Finance Provider configurability, approval controls, auditability, and scalability. This document incorporates the end-to-end business processes, integration requirements, underwriting controls, Finance Provider engagement models, multi-tenant operating principles, and compliance requirements required to support the platform.

1.1 Primary Objectives

- Approval TAT < 24 hours for standard salaried applications (lead → LPO generation).
- Fully digital, no-branch-visit customer journey with Shariah-compliant Diminishing Musharakah structure.
- Automated underwriting via Bank Statement AI parsing, Salary OCR, and BRE decisioning.
- Mandatory Maker-Checker manual review for all customer-uploaded documents (compliance policy).
- QCB-compliant audit logging, PDPPL data residency (Qatar Cloud based), 10-year WORM retention.
- Asset-backed financing only — LPO issued to dealer; zero direct cash disbursement to customer.

1.2 In-Scope Capabilities

- Digital Onboarding & D-KYC (QID OCR, Face liveness, OTP, AML/PEP/Sanctions screening)
- Credit Assessment & BRE (Bank Statement AI parsing, Salary OCR, DBR/FOIR, risk scoring)
- Manual Underwriting with Maker-Checker (Case Manager, CPV, document verification, Sharia audit)
- Offer & Contract Generation (Diminishing Musharakah offer, PDF contracts, manual signature workflow)
- LMS Integration & LPO Dispatch (synchronous LMS booking, asynchronous LPO to dealer)
- Reporting, Audit & Compliance (QCB regulatory extracts, WORM audit logs, PDPPL, DIMINISHING MUSHARAKAH oversight)
- Finance Provider configurability (NBFC Al Jazeera Finance; BLOX)

1.3 Explicitly Out of Scope

Excluded Capability	Owner / Reason
BLOX Admin Portal — Dealer & Inventory Management (URL: https://panel.blox-it.com/admin/auth/login)	BLOX Engineering
Customer-facing vehicle browsing & dealer marketplace (URL: https://www.blox-it.com/customer/auth/login)	Blox
Physical funds transfer to dealer	External settlement system downstream of LPO
Mandate registration processing (execution)	Captured as data; execution is downstream
Cash / cheque disbursements	Strictly LPO-only model

1.4 Product Variants Covered

Variant	Rental Rate	Max Financing (QAR)	Max Tenure
New Car — Standard (<=QAR 90k)	5% to 15%	50,000	60 months (Qatari) / 48 months (Expat)
New Car — Premium (>QAR 90k)	5% to 15%	70,000	60 months (Qatari) / 48 months (Expat)
Used Car	5% to 20%	50,000	48 months
Motorcycle (Used, New)		15,000	

1.5 Product Variant Approval Authority Matrix

The platform shall support the following product-specific financing limits and approval authority thresholds.

Product Variant	Currency	Maximum Finance Amount	Senior Manager – Credit Risk Authority	Head of Credit Risk Authority
Blox Auto Financing (Car)	QAR	70,000	Up to 50,000	Up to 70,000
Blox Auto Financing (Motorcycle)	QAR	15,000	Up to 10,000	Up to 15,000

The approval authority matrix shall be maintained as a configurable parameter and shall be used by the Underwriting Service to determine the required approval level for each application.

2. Scope of This Document

This section defines the boundary of the LOS platform for the purposes of the microservice architecture introduced in Section 5. It is additive to, and consistent with, Sections 1.2–1.3 above.

2.1 Architecture Scope Statement

This document covers the microservices developed and operated within the M2P LOS Platform. Systems outside this boundary are treated as external, integrated systems with defined data contracts (Section 5.7), and their internal implementation is out of scope.

2.2 External Systems (Out of LOS Platform Boundary)

External System	Nature	Relationship to LOS
Blox Customer Portal	External system — customer/dealer-facing UI, owned by BLOX Engineering	Originates applications; consumes LOS status, offers and agreements via API
M2P LMS (Loan Management System)	External, downstream core system	Receives booking/disbursement payload from LOS; returns account/booking status
KYC Platform (D-KYC / AML vendor)	External, third-party verification platform	Performs end-to-end KYC and AML verification and returns verified customer information, documents, and screening outcomes to LOS
Bank Statement AI Vendor	External analytics vendor	Parses bank statements asynchronously via webhook callback
Payment Gateway	Payment gateway integration	Processes customer down payment
BLOX Communication Gateway	External/BLOX-owned messaging middleware	Delivers SMS / WhatsApp / Email notifications

External System	Nature	Relationship to LOS
Finance Provider (NBFC), Model B only	External, human-mediated recipient — not a system integration	Receives the Credit File (Section 5.8) via BLOX Ops outside the LOS platform boundary; no API contract, no automated data flow

3. Key Assumptions

3.1 Functional Assumptions

- One of the Finance Provider is Al Jazeera Finance (NBFC); post-license = BLOX will also be one of the Finance Providers.
- All integrations (D-KYC, AML, Bank Statement Analysis) live and contract-signed before UAT.
- Diminishing Musharakah is the sole Sharia financing structure.
- No STP auto-approval exists for each customer financing application — Maker-Checker is mandatory per compliance policy.
- All PII, SFI, and biometric data stored exclusively within Qatar-hosted cloud environment.
- Vehicle is asset-backed; LPO issued to dealer. Zero direct cash disbursement to customer.
- DIMINISHING MUSHARAKAH must sign off any logic changes to SHARIA001 or fee routing before PROD deployment.
- WPS API provided via BLOX Middleware — M2P calls BLOX internal endpoint.

4. Pre-Requisites Before Go-Live

#	Pre-Requisite	Owner	Gate Type
P1	DIMINISHING MUSHARAKAH Sharia Board sign-off on DM logic & contracts	BLOX Sharia	Hard Gate — no PROD without
P2	Qatar hosted cloud (UAT/PROD) provisioned + data residency confirmation	BLOX DevOps	Hard Gate
P3	D-KYC vendor API credentials + SLA contract	M2P Integrations	Required for DEV
P4	AML/PEP vendor — API key + watchlist subscription	M2P Integrations	Required for UAT
P5	Bank Statement Analysis API — credentials + accuracy SLA	M2P Integrations	Required for UAT
P7	LMS API contract + integration spec (Pre License Phase)	BLOX Technology	Required for UAT
P8	BRE parameters configured (DBR caps, age limits, rate cards)	BLOX Credit Risk	Required for UAT
P9	Vehicle, Employer, Dealer Master data loaded	BLOX Ops	Required for UAT
P10	BLOX Communication Gateway (SMS/WA/Email) credentials	BLOX Technology	Required for UAT

5. Microservice-Based Architecture

Solution Architecture Overview

The LOS platform is composed of seven LOS-owned microservices operating within a single platform boundary (Section 2), integrating with three external systems: the Blox Portal (customer/dealer-facing UI), the KYC Platform (identity verification and AML/PEP/Sanctions vendor), and M2P LMS (downstream loan management system for booking and disbursement).

- Five of the Six LOS microservices are workflow-owning services, each responsible for one stage (or part of a stage) of the existing business journey (Section 8): Customer & Application Creation, KYC Integration, Underwriting, Offer Management, Legal Agreement, and Disbursement Services.
- The seventh, the BRE Service, is a shared, cross-cutting microservice with no workflow stage of its own — it is invoked by the workflow-owning services at defined decision points (Section 6, BRE Architecture / BRE Invocation Matrix).
- The **Blox Portal** originates every application and consumes status/offer/agreement data back from LOS; it does not sit inside the LOS platform, and **its internal implementation is out of scope of this document**.
- The KYC Platform performs identity verification and screening on behalf of the KYC Integration Service and returns verified results to M2P LOS.
- M2P LMS is the system of record after disbursement; the Disbursement Service hands off a booking payload and receives back an account/booking reference, after which the application exits the LOS platform boundary.

5.1 Architectural Principles

- Domain-oriented decomposition. Each LOS microservice owns a coherent business capability (customer/application intake, KYC, underwriting, offer & legal agreement, disbursement) with its own data and API surface.
- BRE is a shared microservice and not a workflow stage. It is invoked synchronously by LOS services whenever business rules, routing decisions, eligibility checks, policy validations, or approval decisioning are required. Multiple rule sets may be configured and executed at different stages of the application lifecycle, including process routing, underwriting routing, and underwriting decisioning. BRE is therefore consumed by multiple LOS services and is not owned by any single process stage.
- Blox Customer Portal, KYC Platform, and M2P LMS are external integrated systems and are not part of the LOS microservice landscape. M2P LOS consumes and exchanges data with these systems through approved integration contracts but does not own or manage their internal data models, business rules, or processing logic (refer Section 5.7).
- Stage-based workflow (Section 8) and service-based architecture (this section) are two views of the same functionality. No requirement, flow, field, API, or status defined elsewhere in this document is altered by this architectural view; this section only adds ownership and integration structure.
- Every existing status, hard gate, and Maker-Checker control in the stage-based workflow remains enforced at the owning service's boundary.

5.2 LOS Microservice Catalog

#	Microservice	Nature	Primary Business Capability
1	Customer & Application Creation Service	LOS-owned (workflow)	Customer identification/creation, application intake, de-

#	Microservice	Nature	Primary Business Capability
			duplication, document & consent capture
2	BRE Service	LOS-owned (shared, cross-cutting)	Eligibility, credit policy, DBR, Sharia (SHARIA001) and exception-routing decisioning, invoked by other services
3	KYC Integration Service	LOS-owned (workflow)	D-KYC orchestration, AML/PEP/Sanctions screening, verification document storage
4	Underwriting Service	LOS-owned (workflow)	Bank statement/salary OCR consumption, manual review, CPV, Maker-Checker credit decision
5	Offer Management & Legal Agreement Service	LOS-owned (workflow)	Offer generation, Musharakah disclosure, customer acceptance Contract bundle generation, signature capture, down payment & repayment mandate, pre-disbursal gate
6	Disbursement Service	LOS-owned (workflow)	Final Ops checklist, LMS booking, LPO generation & dispatch

5.3 Service Responsibilities & Boundaries

Each subsection below states what the service is responsible for, which existing FSD stage(s)/requirement(s) it owns, and its explicit boundary (what it does NOT own).

Service Responsibility Matrix

Quick-reference matrix; full detail for each service follows in Sections 5.3.1–5.3.7 below.

Microservice	Core Responsibility	Owens (FSD Stage)	Boundary (Does Not Own)
Customer & Application Service	Customer ID/creation, de-duplication, application intake, document & consent capture	Stage 1	Identity verification, credit decisioning, document forensics
BRE Service (shared)	Eligibility, credit policy, DBR, Sharia schedule, exception-routing decisioning — invoked, not sequenced	None (shared, cross-cutting — see Section 6)	Data persistence, external vendor calls, customer-facing UI
KYC Integration Service	D-KYC orchestration, AML/PEP/Sanctions screening, verification document storage	Stage 2	Underwriting/credit decisioning

Microservice	Core Responsibility	Owens (FSD Stage)	Boundary (Does Not Own)
Underwriting Service	BSA/salary OCR consumption, manual review, CPV, Maker-Checker credit decision	Stages 3–4	Offer generation, rule evaluation (delegated to BRE)
Offer & Legal Agreement Management Service	Offer generation, Musharakah disclosure, customer acceptance Contract bundle generation & signing, down payment/mandate, pre-disbursal gate	Stage 5, Part 1 & 2	Contract bundle generation, down payment/mandate Final Ops checklist, LMS booking
Disbursement Service	Ops final checklist, LMS booking, LPO generation & dispatch	Stage 6	Contract content changes

5.3.1 Customer & Application Creation Service

Owens: Stage 1 — Onboarding & Application (Section 8, Stage 1).

- Receive application payload from Blox Portal; customer identification via Qatar ID or Expatriate ID.
- Customer creation, de-duplication, and active-application control (see Scenarios A–D below).
- Application creation, vehicle/financing parameter capture, status management, data persistence.
- Document submission intake and consent capture (Qatar ID, Salary Certificate, Bank Statements, Credit Bureau Report, Passport where applicable).
- Invoke BRE Service to determine whether D-KYC is required for the application.

Customer Identification & De-Duplication Scenarios

Scenario	Condition	Service Behaviour
A	Customer Not Found	Create new Customer ID and Application ID
B	Active Application Exists (Created, Submitted, KYC Pending/In Progress, Underwriting Pending, Under Review, Decision Pending)	Do not create new application; return existing Application ID and status
C	Previous Application Approved	Reuse Customer ID; create new Application ID
D	Previous Application Rejected	Reuse Customer ID; create new Application ID

Test Scenarios

Scenario	Expected Result
New applicant double-submits (duplicate click) within seconds	Scenario B applies; same Application ID and status returned; no duplicate record created
Customer with an existing KYC Pending/In Progress application starts a new application	Existing Application ID and status returned; new application creation blocked
Customer with a prior REJECTED application reapplies	Scenario D applies; new Application ID created under same Customer ID; prior rejection reason not disclosed to customer

Scenario	Expected Result
QID on new submission matches an existing customer but name/date of birth differs	Flagged MISMATCHED_IDENTITY; held for manual identity reconciliation by Ops before Scenarios A–D are applied
Customer with a previously APPROVED but undisbursed application starts a new application	Scenario C applies; new Application ID created under same Customer ID; both applications tracked independently

5.3.2 BRE Service (Shared Microservice)

BRE Service (Shared Microservice)

Purpose:

Provides centralized business rule execution, routing decisions, eligibility assessment, policy validation, underwriting decisioning, and Finance Provider-specific workflow determination across the LOS platform.

Responsibilities:

- Execute configurable business rules invoked by LOS services at various stages of the application lifecycle.
- Determine process routing decisions, including whether KYC is required and the applicable customer journey.
- Evaluate Finance Provider-specific rules to determine the underwriting ownership model and subsequent processing path.
- Determine whether underwriting will be performed within M2P LOS or through an external Finance Provider workflow.
- Execute underwriting rule sets for eligibility validation, policy assessment, risk evaluation, vehicle assessment, and product eligibility.
- Support approval decisioning, rejection decisioning, exception handling, referral routing, and manual review determination.
- Enable Finance Provider-specific, product-specific, and process-specific rule configuration without requiring application code changes.
- Provide a centralized rule management framework to ensure consistent decisioning across all LOS services.

Rule Categories:

- Process Routing Rules
- KYC Routing Rules
- Underwriting Routing Rules
- Finance Provider-Specific Rules
- Underwriting Decision Rules
- Approval / Rejection Rules
- Exception & Referral Rules

Typical Invocations:

Invoking Service	Purpose
Customer & Application Service	Determine KYC requirement and applicable application journey

KYC Integration Service	Determine underwriting route and ownership
Underwriting Service	Execute eligibility, policy, risk, approval, rejection, referral, and exception rules

Boundary:

- BRE does not create or maintain customer records, application records, KYC records, agreements, offers, or disbursement records.
- BRE does not perform customer verification activities such as OCR, liveness checks, face matching, AML screening, sanctions screening, or watchlist screening.
- BRE does not directly interact with external customer-facing channels.
- BRE operates solely as a centralized rule evaluation engine and returns decision outcomes to the invoking LOS service, which remains responsible for executing the resulting business action.

5.3.3 KYC Integration Service

Owens: Stage 2 — D-KYC Verification (Section 8, Stage 2).

- KYC initiation, tracking, result consumption, and AML/PEP/Sanctions management.
- Generate KYC Reference Number and KYC Link/URL, shared back to Blox Portal.
- Orchestrate D-KYC activities at the vendor: Consent, Qatar ID Front/Back or Passport capture, Selfie Holding ID, Face Selfie, OCR Extraction, Liveness Detection, Face Match, AML Screening (Sanctions, PEP, Local and International Watchlists).
- Consume vendor results: status, OCR data, AML result, Face Match score, Liveness result, images, documents, reference number.
- Store KYC documents, AML reports, verification results and audit trail.
- Invoke BRE Service to determine the underwriting route once verification is complete.

Boundary: does not perform underwriting or credit decisioning; hands off a verified/eligible customer to the Underwriting Service via BRE routing.

5.3.4 Underwriting Service

Owens: Stage 3 — Bank Statement Assessment, Salary Certificate OCR , and Stage 4 — Approval or Decisioning (Case Manager) (Section 8, Stages 3–4).

- Bank statement ingestion, AI parsing, and tamper-detection escalation.
- Bureau report OCR parsing or Qatar Bureau integration
- Manual verification of Bureaus' parsed data, Salary certificate, income verification, and income cross-verification against bank statements.
- Submit inputs to BRE Service for credit policy decisioning (eligibility, credit bureau details, DBR/stress test, Sharia structure, exception routing) and consume its decision.
- Contact Point Verification (CPV) flow if required based on Credit Underwriter's decision
- Maker-Checker credit decision workflow — approve, reject, rework, or pend.
- Compile the Credit File — application data, customer documents, KYC results, bank/bureau analysis, and BRE output — for Finance Providers configured under Credit File Handoff (Section 5.8, Model B).
- For Model B Finance Providers, capture the Finance Provider's underwriting decision — made outside LOS — back into the application (Section 5.8).

External Decision Capture (Model B): because the Finance Provider's decision is made outside LOS, it is entered under the same two-person integrity control as the in-LOS Maker-Checker workflow (Section 8, Stage 4): a first OPS_ADMIN records the Finance Provider's decision (APPROVE/REJECT/PEND) against the Finance Provider's written

communication, tagged decision_source = EXTERNAL_FINANCE_PROVIDER; a second OPS_ADMIN or FIN_OPS user who did not make the original entry confirms it before status advances. The same-user restriction in Stage 4's Separation of Duties rule applies identically here. This is the only mechanism by which a Model B application reaches FINANCING_APPROVED or REJECTED_BY_UNDERWRITER.

Boundary: does not generate the customer-facing offer (owned by Offer Management Service); does not evaluate rules itself — all policy evaluation is delegated to the BRE Service.

5.3.4.1 Product Variant Assessment and Approval Authority Validation

The Underwriting Service shall determine the applicable product variant based on application details, vehicle category, Finance Provider configuration and selected financing product.

Prior to final decisioning, the service shall validate that the requested financing amount does not exceed the maximum permissible financing amount configured for the selected product variant (Section 1.5).

The Underwriting Service shall determine the required approval authority using:

- Product Variant
- Requested Finance Amount
- Underwriting Outcome
- Applicable Approval Matrix (Section 1.5)

Where the requested finance amount exceeds the approval authority of the currently assigned approver, the application shall be automatically escalated to the next approval level.

Approval authority thresholds shall be maintained through configuration and shall support future changes without application code modification.

The determined approval level, approval routing outcome and escalation history shall be stored as part of the underwriting audit trail and made available to maker-checker workflow, reporting and audit functions.

5.3.5.1 Offer Management Service

Owns: Stage 5, Part 1 — Offer Generation, Review & Customer Acceptance (Section 8, Stage 5).

- Generate the financing offer including full Musharakah breakdown (Asset Value, BLOX%/Customer% share, Profit Rate, Rent per period, Total Financing Cost, Amortization Schedule).
- Send the offer/LPO-eligible terms to Blox Portal for customer review.
- Receive customer acceptance/rejection and the signed LPO where applicable, and store the outcome.
- Enforce the pre-contract Sharia disclosure hard gate before the acceptance action is enabled.

Boundary: does not generate the legal contract bundle itself (owned by Legal Agreement Service) and does not manage down payment or repayment mandate collection.

5.3.5.2 Legal Agreement Service

Owns: Stage 5, Part 2 — Contract Generation, Signing, Verification & Pre-Disbursal Setup (Section 8, Stage 5).

- Generate the contract based on template provided by Finance Provider [eg: Diminishing Musharakah Agreement and full contract bundle (Agency Agreement/Wakala, Key Facts Statement, Credit Approval Memo, Wa'ad, takaful declaration)]
- Send agreement(s) to Blox customer Portal and receive signed documents; verify and store executed agreements.

Boundary: does not perform the final Ops checklist, VIN/takaful confirmation, or LMS booking (owned by Disbursement Service).

5.3.6 Disbursement Service

Owns: Stage 6 — Final Ops Review (Section 8, Stage 6).

- Run the Ops Final Checklist (contracts signed & verified, down payment receipted, repayment mode registered, VIN/Chassis/Engine confirmed, takaful active).
- Validate the signed agreement and mark the application as Disbursed.
- Generate the LMS booking payload and perform the synchronous LMS integration call.

Boundary: does not modify contract content (owned by Legal Agreement Service); LMS itself is external and out of the LOS platform boundary (Section 2).

5.4 Service Ownership Boundaries

Organizational and data ownership per service, consistent with the Shared Responsibility Matrix (Section 12) and RBAC roles (Section 7).

Service	Technical Owner	Business/Policy Owner	Primary Data Owned	Primary Roles Interacting
Customer & Application Creation Service	M2P	BLOX Admin BLOX Ops Ops (Finance Providers)	Customer profile, Lead, Application header	CUST, OPS_ADMIN, CASE_MGR
BRE Service	M2P	BLOX Admin BLOX Ops Ops (Finance Providers) BLOX Credit Risk / DIMINISHING MUSHARAKAH (for SHARIA001)	bre_parameters, bre_execution_log	SYS_ADMIN (config), CRCH_UW/CRCH_CHK (consumers)
KYC Integration Service	M2P	BLOX Compliance / MLRO BLOX Admin BLOX Ops Ops (Finance Providers)	KYC documents, AML/PEP results, verification audit trail	CASE_MGR, AUDIT_COMP
Underwriting Service	M2P	BLOX Credit Risk Finance Provider's Credit Risk	Case queue, BSA/salary extraction, underwriting decision, Credit File (Model B, Section 5.8)	CRCH_UW, CRCH_CHK, BLOX_RISK
Offer & Legal Agreement Management Service	M2P	BLOX Ops / Finance Provider's Ops	Offer, amortization schedule	CUST, OPS_ADMIN
Offer & Legal Agreement Management Service	M2P	BLOX Ops / Finance Provider's Ops	Contract bundle, signed agreements, down payment/mandate record	CUST, OPS_ADMIN

Service	Technical Owner	Business/Policy Owner	Primary Data Owned	Primary Roles Interacting
Disbursement Service	M2P	BLOX Ops Finance Provider's Ops	Disbursement record, LPO, LMS booking reference	FIN_OPS

Integration Landscape

5.5 End-to-End Service Integration Flow

This is the sequence in which services are invoked for a standard application, exactly as defined in the Qatar Vehicle Finance detail section. It is a service-level view of the same journey described stage-by-stage in Section 8.

Flow — Model A (Full LOS Underwriting): Blox customer Portal → Customer & Application Creation Service → BRE → KYC Integration Service → BRE → Underwriting Service → BRE → in-LOS Maker-Checker (Section 8, Stage 4) → Offer Management Service → Legal Agreement Service → Disbursement Service → M2P LMS

Flow — Model B (Credit File Handoff, Section 5.8): Blox customer Portal → Customer & Application Creation Service → BRE → KYC Integration Service → BRE → Underwriting Service → BRE → Credit File compiled; BLOX Ops hands off to Finance Provider outside the LOS boundary → BLOX Ops records external decision (Section 5.3.4) → Offer Management Service → Legal Agreement Service → Disbursement Service → M2P LMS

Both flows are identical through the third BRE invocation. Which flow a given application follows is fixed at the Finance Provider level by Engagement Mode (Section 5.8) — it is never a case-by-case choice.

5.6 Inter-Service Integration Points (Integration Matrix)

From Service	To Service	Trigger	Sync/Async	Purpose
Customer & Application Service	BRE Service	After application creation	Synchronous	Determine whether D-KYC is required
Customer & Application Service	KYC Integration Service	BRE routes "KYC Required"	Synchronous (handoff), async verification	Begin D-KYC verification
KYC Integration Service	BRE Service	After D-KYC + AML result available	Synchronous	Determine underwriting route / eligibility gate
KYC Integration Service	Underwriting Service	BRE confirms eligibility	Synchronous (handoff)	Begin credit assessment
Underwriting Service	BRE Service	After BSA/Salary OCR/Credit Bureau inputs collected	Synchronous	Eligibility, policy, risk, exception-tier decisioning
Underwriting Service	Offer Management Service	Maker-Checker decision = APPROVE (FINANCING_APPROVED)	Synchronous (handoff)	Generate financing offer
Offer Management Service	Legal Agreement Service	Customer accepts offer	Synchronous (handoff)	Generate contract bundle

From Service	To Service	Trigger	Sync/Async	Purpose
Legal Agreement Service	Disbursement Service	Pre-Disbursal Completion Gate satisfied	Synchronous (handoff)	Run final Ops checklist and LMS booking
Disbursement Service	M2P LMS (external)	Ops checklist passed	Synchronous	Create financing contract / account in LMS
Disbursement Service	Blox Portal / Dealer (external, via BLOX Middleware)	LMS booking success	Asynchronous	Dispatch signed LPO
Underwriting Service	BLOX Ops (human-mediated, Model B only)	BRE-based eligibility check complete AND Finance Provider Engagement Mode = Credit_File_Handoff (Section 5.8)	Synchronous compilation; asynchronous, human-mediated handoff	Compile Credit File and bypass in-LOS Maker-Checker; external decision captured per Section 5.3.4

Timeout, retry and fallback behaviour for each of these hops is governed by the existing Integration Matrix and Universal Integration Rules (Section 9.3–9.4), which apply equally whether the call is service-to-vendor or service-to-service.

5.7 Data Exchange Contracts

This subsection satisfies the requirement to define the data exchanged (a) between Blox Portal and LOS services, (b) between LOS services and the KYC platform, and (c) between LOS services and the LMS. Field-level detail already defined elsewhere in this document (Sections 8, 9) is referenced rather than restated in full.

5.7.1 Blox Portal ↔ LOS Services

Direction	Data	Owning LOS Service
Portal → LOS	Customer information (name, QID/Passport, contact, nationality)	Customer & Application Service
Portal → LOS	Vehicle information (price, status, make, model, year)	Customer & Application Service
Portal → LOS	Financing information (down payment, tenure, salary, liabilities, employment type)	Customer & Application Service
Portal → LOS	Dealer's details (Dealer name, branch, executive name, executive Unique Id)	Customer & Application Service
Portal → LOS	Finance Provider's detail (Finance Provider name, Branch, Finance Provider unique id)	Customer & Application Service
Portal → LOS	Supporting documents (QID, Salary Certificate, Bank Statements, Credit Bureau Report, Passport, Residence proof, Additional docs)	Customer & Application Service
Portal → LOS	Consents (Credit Bureau, T&C, D-KYC/Biometric, AML)	Customer & Application Service
Portal → LOS	Offer acceptance / rejection response	Offer Management Service
Portal → LOS	Signed contract document uploads	Legal Agreement Service
Portal → LOS	Down payment initiation, repayment mandate selection	Legal Agreement Service
LOS → Portal	Application ID, Customer ID, Status, KYC URL/reference	Customer & Application Service / KYC Integration Service

Direction	Data	Owning LOS Service
LOS → Portal	Financing offer (LPO terms, Musharakah breakdown, amortization schedule)	Offer Management Service
LOS → Portal	Agreement documents for signature	Legal Agreement Service
LOS → Portal / Dealer	Signed LPO, disbursement confirmation	Disbursement Service
LOS → Portal	Real-time application status updates (all stages)	Customer & Application Service (status), all services (source of truth per stage)

5.7.2 LOS Services ↔ KYC Platform

Direction	Data	Owning LOS Service
LOS → KYC Platform	Consent flag, Qatar ID/Passport images, Selfie-holding-ID image, Face Selfie	KYC Integration Service
KYC Platform → LOS	KYC Reference Number, verification Status, OCR-extracted data, Face Match score, Liveness result	KYC Integration Service
KYC Platform → LOS	AML/PEP/Sanctions screening result (CLEAR / POTENTIAL_MATCH / CONFIRMED_MATCH) and confidence	KYC Integration Service
KYC Platform → LOS	Verification images/documents for storage, plus vendor audit trail reference	KYC Integration Service

Endpoint-level specification (POST /api/initiate-dkyc, POST /api/aml/screen, timeouts, retries, thresholds) is unchanged and defined in Section 8, Stage 2.

5.7.3 LOS Services ↔ LMS

Direction	Data	Owning LOS Service
LOS → LMS	Customer data, Facility/financing data, Vehicle data, Agreement data, Booking/contract-creation data	Disbursement Service
LMS → LOS	LMS Reference Number / Account Number	Disbursement Service

Final status on success: Application = Disbursed, Facility = Booked. Endpoint-level specification (POST {LMS_BASE_URL}/api/contracts/create, 30s timeout, idempotency key, rollback-on-failure behaviour) is unchanged and defined in Section 8, Stage 6.

5.8 Deal-Based Finance Provider Engagement Model

The extent to which a given Finance Provider (NBFC) uses M2P LOS is not uniform across Finance Providers on the platform. It is determined by the commercial and operational deal signed between BLOX and that Finance Provider — configured per Finance Provider in the Finance Provider Master (Section 7.1, Finance Provider’s Input & Policy Master) and evaluated by BRE’s Finance Provider-specific rules (Section 5.3.2, Section 6.3) at the point process routing is decided. This determines which LOS services a given Finance Provider’s applications actually pass through, and whether that Finance Provider’s own team ever accesses M2P LOS directly.

EXPLICIT ENGAGEMENT DETERMINATION: Which M2P LOS flows a Finance Provider uses is set by its signed deal with BLOX — it is never assumed to be uniform across Finance Providers. For example, a given NBFC may never use M2P LOS for underwriting at all, while still requiring a baseline BRE eligibility/policy check on every application (performed using BLOX’s BRE, or a combined BLOX + that NBFC’s BRE). This distinction is configured explicitly per Finance Provider before go-live (see Configuration Attributes below) — it is never inferred from usage patterns.

Engagement Model	Underwriting Ownership	M2P LOS Services Invoked	Finance Provider's Direct LOS Access
Model A — Full LOS Underwriting	Finance Provider's own Credit Risk team, working inside M2P LOS	All of Stages 1–6, including the Underwriting Service's Maker-Checker queue (Section 8, Stage 4)	Yes — the Finance Provider's underwriting users are provisioned into LOS, scoped to their own applications only (Section 7.4)
Model B — Credit File Handoff (Underwriting Outside LOS)	Finance Provider's own Credit Risk team, working outside M2P LOS	Stages 1–3 (intake, D-KYC, BRE-based eligibility check) run inside LOS; Stage 4 Maker-Checker is NOT executed in LOS for this Finance Provider — LOS instead compiles a Credit File (below)	No — the Finance Provider does not log into M2P LOS; BLOX Ops interacts with LOS on the Finance Provider's behalf

Example — NBFC A: Under its commercial agreement with BLOX, NBFC A never accesses M2P LOS for underwriting. It requires only that a baseline BRE eligibility/policy check be performed on the application — this may be BLOX's standard BRE rule set, or a combined BLOX + NBFC A rule set configured specifically for NBFC A (Section 5.3.2, Rule Categories: Finance Provider-Specific Rules). NBFC A is therefore configured as Model B (Credit File Handoff).

Credit File — Contents & Process (Model B Only)

For any Finance Provider configured under Model B, M2P LOS prepares a Credit File once BRE-based processing on the application is complete. The Credit File is a system-generated bundle, not an ad hoc export, and contains:

- Application data (customer, vehicle, and financing parameters captured at Stage 1)
- Documents submitted by the customer (Qatar ID, Salary Certificate, Bank Statements, Credit Bureau Report, Passport where applicable — Section 8, Stage 1)
- KYC verification results (D-KYC and AML/PEP/Sanctions outcome — Section 8, Stage 2)
- Bank statement and credit bureau analysis [data parsing] (Section 8, Stage 3)
- The underwriting BRE output as defined in the system (eligibility, DBR, risk tier, and any exception-routing recommendation — Section 6)

BLOX Ops (OPS_ADMIN) downloads the Credit File and shares it with the Finance Provider outside the M2P LOS system boundary. This is stated explicitly because it is the one point in the journey where a Finance Provider-scoped application leaves the platform through an operational, human-mediated step rather than an API integration — the file transfer itself sits outside LOS's system boundary (Section 1.3), in the same way physical fund transfer and mandate execution do.

Because the Finance Provider underwrites offline under Model B, its decision is not produced by the in-LOS Maker-Checker workflow. The Underwriting Service captures this external decision under a two-person integrity control (Section 5.3.4, External Decision Capture) and tags it `decision_source = EXTERNAL_FINANCE_PROVIDER`. Once confirmed, the application proceeds to Offer Generation (Stage 5) exactly as it would under Model A; while awaiting the Finance Provider's decision, the application sits at status `PENDING_EXTERNAL_DECISION` (Application Status Registry, Section 8).

Reports and Module Visibility Under Model B: whether a Finance Provider's applications are processed under Model A or Model B, BLOX retains full visibility of its own platform. Blox Admin and Blox Ops continue to see the full Reporting & MIS Dashboard, Audit Trail, and all other Admin Modules (Section 7.1) for every Finance Provider relationship, including Model B Finance Providers who never log into LOS themselves. Model B does not reduce BLOX's own visibility — it only removes the Finance Provider's own team from directly using the LOS underwriting workflow (see Section 7.4, Multi-Tenant Data Segregation, for the full visibility model).

Configuration Attributes (Finance Provider Master)

Attribute	Values	Set By	Effect
Engagement Mode	Full_LOS_Underwriting / Credit_File_Handoff	BLOX Admin (SYS_ADMIN), per signed Finance Provider agreement	Determines whether Stage 4 Maker-Checker executes inside LOS (Model A) or a Credit File is compiled instead (Model B)
BRE Ownership	Blox_BRE_Only / Blox_Plus_Finance Provider_BRE	BLOX Admin (SYS_ADMIN), with BRE Finance Provider- specific rule configuration (Section 5.3.2)	Determines whether the baseline BRE check uses BLOX's standard rule set alone, or a combined BLOX + Finance Provider rule set

Turnaround Targets (Model B): Credit File compilation completes within 30 minutes of the Stage 3 BRE check passing (system-generated; no manual step). Recording of the Finance Provider's external decision back into LOS is targeted within 1 business day of receipt by BLOX Ops; cases exceeding this trigger the same SLA-breach alerting as Underwriting Pending cases (Section 10).

Test Scenarios

Type	Scenario	Expected Output
Happy	Finance Provider configured Model A; application reaches Stage 4	Maker-Checker queue in LOS shows the case to the Finance Provider's CRCH_UW; standard Section 8, Stage 4 flow applies
Happy	Finance Provider configured Model B (e.g., NBFC A); application completes the Stage 3 BRE check	Stage 4 in-LOS Maker-Checker queue is bypassed; Credit File is compiled and made available to OPS_ADMIN for download
Negative	A Model B Finance Provider's case is mistakenly routed to the in-LOS Underwriting Queue	Blocked by the BRE process-routing rule; case routed to Credit File compilation instead; configuration-mismatch alert raised to SYS_ADMIN
Negative	A non-Ops role (e.g., CRCH_UW) attempts to download a Credit File	403 — Credit File download restricted to OPS_ADMIN/FIN_OPS; attempt logged
Edge	Engagement Mode changed mid-application (Model A → Model B)	Not permitted on an in-flight application; the change applies only to new applications under the affected Finance Provider, from the effective date

6. BRE Architecture

This is a new, dedicated architecture section for the Business Rule Engine (BRE), separate from the operational BRE Governance content in Section 9 (Fee Master & Parameter Dictionary, which is preserved unchanged). It exists to make explicit, at an architectural level, how the BRE is built, invoked and governed as a shared microservice.

6.2 BRE Invocation Matrix

Invoking Service	Trigger Point	BRE Decision Returned
Customer & Application Service	Immediately after application creation	Is D-KYC required for this application? Additionally returns the Finance Provider's Engagement Mode (Full_LOS_Underwriting / Credit_File_Handoff) and BRE Ownership (Section 5.8), read from the Finance Provider Master and evaluated once per application. This value is cached on the application record and referenced — not re-evaluated — at every later stage.
KYC Integration Service	After D-KYC and AML/PEP/Sanctions results are available	Determine the underwriting route; confirm residency/initial eligibility (ELIG001 Phase 1)
Underwriting Service	After Bank Statement AI parsing, Salary OCR and Credit Bureau data are collected	Eligibility & knockout (ELIG001), Credit Bureau evaluation (CBS001), DBR & stress test (DBR001), Sharia schedule computation (SHARIA001), exception routing (EXC001), and auto vs. manual decisioning

These three invocation points correspond exactly to the three appearances of BRE in the End-to-End Service Integration Flow (Section 5.5).

6.3 BRE Rule Domains (Reference)

The rule domains themselves — including pseudo-code, thresholds and exception routing logic — are business logic owned by BRE Governance (Section 9) and are not modified here. They are indexed below purely to connect them to their invocation point in the architecture.

Rule Domain	Purpose	Invoked By (Architecture View)
ELIG001 — Eligibility & Knockout	Fraud/sanctions hard-block, age, income, residency, employer-blacklist checks	KYC Integration Service (Phase 1) and Underwriting Service (full evaluation)
CBS001 — QCB Credit Bureau	Bureau score evaluation, thin-file handling, risk tiering	Underwriting Service
DBR001 — Debt Burden Ratio & Stress Test	DBR calculation, exception tiering, stressed-DBR test	Underwriting Service
SHARIA001 — Diminishing Musharakah	Rental/principal schedule computation, late fee (Gharamah) and Ibra rules	Underwriting Service (schedule generation) and Offer Management Service (disclosure)
EXC001 — Exception Routing	Routes exceptions to Head of Credit / CRO / CRO+CEO based on tier	Underwriting Service
FINANCE PROVIDER001 — Engagement Mode Routing	Determines Full LOS Underwriting vs. Credit File Handoff per the Finance Provider Master (Section 5.8)	Customer & Application Service, at application creation

Full pseudo-code, thresholds and configurable parameters for each rule domain are unchanged and remain in Section 9 ("BRE Governance — Fee Master & Parameter Dictionary") and within Stage 3 of Section 8.

Note: The Credit Assessment Business Rules are maintained as a configurable module within the system. Authorized administrators can modify these rules as required through the configuration interface without requiring application code changes, provided the changes are limited to existing configurable parameters and predefined values available within the system.

Any modification involving the introduction of a new assessment field, attribute, data element, calculation parameter, or evaluation criterion that is not currently available in the configuration framework shall be treated as a Change Request (CR). Such enhancements will require formal requirement analysis, design review, development, testing, and deployment through the standard software development lifecycle (SDLC) process before being made available in production.

6.4 BRE Service Contract

At a field level (not a binding schema), each BRE invocation follows the same request/response shape:

- Request: invoking_service, application_id, decision_point (e.g. KYC_REQUIRED_CHECK, UNDERWRITING_ROUTE, CREDIT_DECISION), and the relevant input payload (e.g. nationality, age, net income, residency status, QCB score, DBR inputs, tenure, risk tier inputs) as applicable to that decision point.
- Response: status/decision code (e.g. AUTO_DECLINED, PROCEED_WITH_FLAG, REFER_CREDIT, exception tier), reason code (e.g. REJ_AGE, REJ_DBR_EXCEEDED), override flag, and — for SHARIA001 — the computed amortization schedule.
- Every invocation is logged to bre_execution_log (WORM via Kafka → Elasticsearch) with full input/output snapshot, per the existing Engineering Checklist (Section 9).

6.5 BRE Configuration & Change Governance

- The Credit Assessment Business Rules are maintained as a configurable module within the system. Authorized administrators can modify these rules as required through the configuration interface without requiring application code changes, provided the changes are limited to existing configurable parameters and predefined values available within the system.
- Any modification involving the introduction of a new assessment field, attribute, data element, calculation parameter, or evaluation criterion that is not currently available in the configuration framework shall be treated as a Change Request (CR). Such enhancements will require formal requirement analysis, design review, development, testing, and deployment through the standard software development lifecycle (SDLC) process before being made available in production.
- SHARIA001 parameter changes require dual sign-off: Product Maker + DIMINISHING MUSHARAKAH Checker before any PROD change (Sections 3, 4.3, 9).
- Simulation: POST /api/bre/simulate performs a dry-run of a proposed change against the last 10,000 historical applications before it is promoted (Section 9).
- Rollback: single-click via active_version_ptr in bre_config, with automatic alert if the rejection rate spikes more than 10% after a change (Section 9).
- Independent assurance: the Reconciliation Module (Section 15) re-computes BRE inputs on a separate rule script every 15 minutes and halts progression on any mismatch, routing to the Model Risk Review Queue.

Because BRE is shared across three invocation points and multiple owning services, this governance model (config-driven parameters, dual sign-off, simulate/rollback, independent reconciliation) is what allows it to be operated safely as a shared microservice rather than duplicated logic inside each workflow service.

6.6 BRE Availability & Fallback Behaviour

Because BRE is invoked synchronously at three points in the journey (Section 6.2) and no downstream service can render a decision without it, BRE's own availability directly gates application progression at those points. As BRE is a shared internal microservice rather than a third-party vendor, it does not appear in the vendor-facing Integration Matrix (Section 9.3); the behaviour below is the internal-service equivalent for that matrix, using the same circuit-breaker and idempotency principles set out in Section 9.4 (Universal Integration Rules).

Invocation Point	Timeout / Retry	Fallback on Failure or Timeout
Post-application-creation (KYC required check, Section 6.2)	3s timeout; 3× exponential backoff (0.5s, 1s, 2s); circuit breaker after 5 failures/60s	Status → KYC_ROUTING_PENDING_MANUAL; routed to Ops queue for manual KYC-requirement determination
Post-D-KYC (underwriting route, Section 6.2)	Same as above	Status → ROUTING_PENDING_MANUAL; routed to Case Manager queue for manual routing
Post-BSA/Salary/Bureau (credit decisioning, Section 6.2)	5s timeout; same retry/circuit-breaker policy	Status → BRE_DECISION_PENDING_MANUAL; case surfaced to the Underwriter flagged that automated decisioning did not complete; a manual eligibility check is required before any Maker action

7. Functional Modules, User Roles & Access Control Matrix

7.1 Functional Modules, Submodules & User Portals

Portal	Module	Sub Module	Owning Microservice
Customer Portal	Customer Journey/Assisted Journey	Application creation detail	Customer & Application Service
Customer Portal	Customer Journey	KYC / Biometric Verification	KYC Integration Service
Customer Portal	Customer Journey	Application Status Tracker	Customer & Application Service (aggregated)
Customer Portal	Customer Journey	Final Offer Review	Offer Management Service
Customer Portal	Customer Journey	Down Payment & Repayment Mode Capture	Legal Agreement Service
Customer Portal	Customer Journey	Pre-Disbursal Confirmation & Agreement Signing	Legal Agreement Service
LOS Case Manager (Back-Office)	Operations & Underwriting	Application Queue	Underwriting Service
LOS Case Manager (Back-Office)	Operations & Underwriting	Application Detail 360°	Cross-service (Customer & Application Service aggregation)
LOS Case Manager (Back-Office)	Operations & Underwriting	Manual Profile Review	Underwriting Service
LOS Case Manager (Back-Office)	Operations & Underwriting	KYC Review & Override	KYC Integration Service
LOS Case Manager (Back-Office)	Operations & Underwriting	AML / Sanctions / PEP	KYC Integration Service
LOS Case Manager (Back-Office)	Operations & Underwriting	Salary & BSA Analysis	Underwriting Service
LOS Case Manager (Back-Office)	Operations & Underwriting	Credit Bureau Report Review	Underwriting Service
LOS Case Manager (Back-Office)	Operations & Underwriting	BRE Decision Summary	BRE Service (shared) — surfaced via Underwriting Service
LOS Case Manager (Back-Office)	Operations & Underwriting	CPV	Underwriting Service
LOS Case Manager (Back-Office)	Operations & Underwriting	Underwriting Review L1	Underwriting Service
LOS Case Manager (Back-Office)	Operations & Underwriting	Final Approval L2	Underwriting Service
LOS Case Manager (Back-Office)	Operations & Underwriting	Credit File Generation & Download	Underwriting Service
LOS Case Manager (Back-Office)	Operations & Underwriting	Pre-Disbursal Checklist	Disbursement Service

Portal	Module	Sub Module	Owning Microservice
LOS Case Manager (Back-Office)	Operations & Underwriting	Financing Creation	Disbursement Service
Admin Modules	Administration	User, Role & Permission Management	Cross-service (platform-level)
Admin Modules	Administration	Workflow Management	Cross-service (platform-level)
Admin Modules	Administration	BRE Module	BRE Service (shared)
Admin Modules	Administration	Finance Provider's input & Policy Master	BRE Service (shared) / Customer & Application Service
Admin Modules	Administration	Master Data Module	Customer & Application Service
Admin Modules	Administration	Notifications & Communication	Cross-service (platform-level)
Admin Modules	Administration	Reporting & MIS Dashboard	Cross-service (platform-level)
Admin Modules	Administration	Audit Trail & Activity Log	Cross-service (platform-level)

7.2 System Roles

Role ID	Role Name	Portal	Primary Responsibilities
CUST	Customer	Customer Portal (self only)	Initiates application, uploads docs, accepts offer, signs contracts, pays down payment (Outside M2P Los System)
Sales_Executive(Dealer)	Sales Executive	Operates Customer portal on behalf of customer	Initiates application, uploads docs, (Outside M2P Los System)
OPS_ADMIN	Ops Admin	LOS Case Manager (no risk overrides)	D-KYC oversight, BRE config, contract upload verification
CASE_MGR	Case Manager	LOS Case Manager (queues)	SLA monitoring, queue hygiene, pending application routing
CRCH_UW	Credit Analyst (Maker)	LOS Underwriting Queue	Manual underwriting, Sharia audit, DBR validation, CPV, first-level approval
CRCH_CHK	Credit Checker	LOS Checker Queue	Second-level binding approve/reject/pending decision
BLOX_RISK	CRO / Risk Committee	LOS Risk Override Queue	Tier 2/3 exception approval, policy deviation sign-off
FIN_OPS	Finance Ops / LPO Approver	LOS Disbursal Queue	Pre-disbursal checklist, LPO authorization, LMS finance creation trigger
AUDIT_COMP	Auditor / MLRO / Sharia Board	Read-only all artifacts	AML/PEP investigation, Sharia review, QCB audit, fraud escalation

Role ID	Role Name	Portal	Primary Responsibilities
SYS_ADMIN	System Administrator	LOS Admin Panel (full config)	User provisioning, BRE parameter management, system config

7.3 RBAC Key Rules

- Maker (CRCH_UW) cannot be Checker (CRCH_CHK) on the same case — enforced at DB constraint level.
- Tier 2 exceptions: only BLOX_RISK (CRO) can act as Checker.
- Tier 3 exceptions: CRO + CEO dual approval required before status advances.
- AUDIT_COMP has read-only access to all artifacts; can unmask PII (logged immutably).
- AML hard-block profiles cannot be accessed or modified by any role except AUDIT_COMP/MLRO.
- SHARIA001 parameter changes require dual sign-off: Product Maker + DIMINISHING MUSHARAKAH Checker.
- Model B external-decision confirmation: the confirming OPS_ADMIN/FIN_OPS cannot be the same user who made the original EXTERNAL_FINANCE PROVIDER entry — enforced at DB constraint level, identical in mechanism to the Maker≠Checker rule above.
- A Checker's Approval Authority (Senior Manager or Head of Credit Risk, Section 1.5) must be equal to or greater than the case's financing amount for the applicable product variant before a Checker decision is accepted as binding; the assigned Checker's authority tier, any escalation, and the resolving Checker's identity are logged to the audit trail.

7.4 Multi-Tenant Data Segregation (Finance Provider-Scoped Visibility)

M2P LOS is a shared, multi-Finance Provider platform: multiple NBFCs (Finance Providers) may be configured on the same LOS instance, each per its own engagement model (Section 5.8). Regardless of engagement model, every application record carries the Finance Provider ID captured at Stage 1 (Section 8, Stage 1 — Finance Provider Details). This Finance Provider ID is the tenant-scoping key enforced across every LOS microservice and every user-facing module in Section 7.1.

TENANT ISOLATION RULE: A Finance Provider's own users see only applications mapped to that Finance Provider. NBFC A's users can never view NBFC B's applications — or any other Finance Provider's — regardless of role, search method, or direct Application ID reference. BLOX Admin and BLOX Ops are the only actors with visibility across all Finance Providers on the platform.

Actor	Visibility Scope	Notes
Finance Provider's own users (e.g., NBFC A's CRCH_UW / CRCH_CHK and equivalent roles, Section 7.2, where the Finance Provider is configured under Model A, Section 5.8)	Applications where application.Finance Provider_id = the user's assigned Finance Provider ID, only	Cannot view, search, or act on another Finance Provider's applications under any circumstance, including by direct Application ID reference
BLOX Admin (SYS_ADMIN)	All applications, all Finance Providers	Full platform configuration and data visibility
BLOX Ops (OPS_ADMIN, CASE_MGR, FIN_OPS)	All applications, all Finance Providers	Required to operate the Model B Credit File handoff (Section 5.8) and any cross-Finance Provider operations

Actor	Visibility Scope	Notes
BLOX Risk & Compliance (BLOX_RISK, AUDIT_COMP)	All applications, all Finance Providers	Platform-wide exception approval, audit, and regulatory oversight

Enforcement Rules

- Finance Provider-scoping is enforced at the API and database layer (row-level filtering on Finance Provider_id), not by UI filtering alone — consistent with the enforcement principle already applied to the AML hard block (Section 8, Stage 2).
- Every API endpoint that returns application-level data (Section 16.2) applies this filter automatically based on the authenticated user's assigned Finance Provider ID; a Finance Provider-scoped user's token carries no ability to widen this scope.
- A Finance Provider-scoped user attempting to access another Finance Provider's application — whether via UI search or a direct API call with a known Application ID — receives 403 Forbidden, and the attempt is logged to the audit trail (Section 11) as a cross-tenant access attempt.
- The Reporting & MIS Dashboard and Audit Trail modules (Section 7.1) apply the same scope: a Finance Provider's users see only their own portfolio's metrics; BLOX roles see the consolidated, all-Finance Provider view.
- This scoping applies regardless of engagement model (Section 5.8): a Model B Finance Provider that never underwrites in LOS is, in principle, still scoped to its own Finance Provider ID for any module it is later provisioned to view.
- Finance Provider ID assignment to a user is a SYS_ADMIN provisioning action (Section 7.1, User/Role/Permission Management) and is captured in the audit trail; a user cannot self-assign or change their own Finance Provider ID scope.

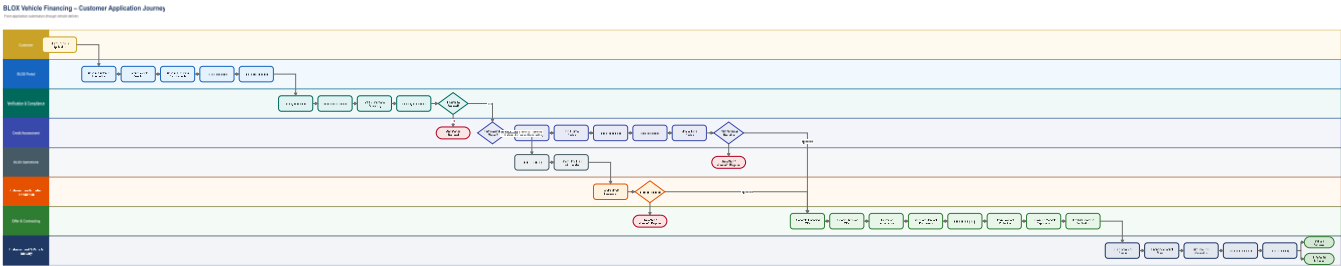
Test Scenarios

Type	Scenario	Expected Output
Happy	NBFC A underwriter calls GET /api/cases/queue	Returns only cases where Finance Provider_id = NBFC A
Happy	BLOX OPS_ADMIN calls GET /api/cases/queue	Returns cases across NBFC A, NBFC B, and every other configured Finance Provider
Negative	NBFC A underwriter calls GET /api/cases/{id} for a known NBFC B application ID	403 Forbidden — 'Application not accessible'; attempt logged
Negative	NBFC A underwriter searches Application Detail 360° by an NBFC B customer's QID	No result returned — not even a 'not found' hint that would distinguish non-existence from access-denial, to avoid leaking NBFC B's customer data
Edge	A user is provisioned against two Finance Providers in error (data entry mistake)	Blocked at provisioning; a Finance Provider-scoped role enforces exactly one Finance Provider ID; multi-Finance Provider assignment requires an explicit BLOX-internal role, not a Finance Provider-side role

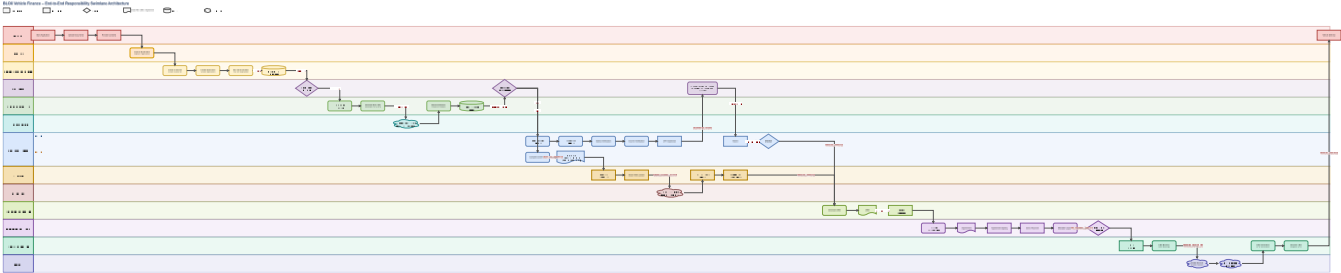
8. Business Workflow (Stage-Based View)

The high-level end-to-end LOS process flow is depicted in the following flow:

Application Journey:



Application and Microservices Interactions (For Developers)



Below is the step-by-step sequence of interactions.

Application Status Registry (Consolidated Reference)

This table consolidates every application.status value introduced stage-by-stage below, including the Model B statuses defined in Section 5.8. It is the authoritative reference for Section 5.3.1’s de-duplication logic (Scenarios A–D) and for every stage’s status-transition language. Sub-flags (e.g., FLAG: income mismatch) that do not change application.status are not repeated here — see the owning stage for those.

Status	Set By (Stage/Service)	Meaning	Active ¹
CREATED	Stage 1	Application record exists, not yet submitted	Yes
SUBMITTED	Stage 1	Payload accepted, pre-dedupe	Yes
KYC_PENDING / KYC_IN_PROGRESS	Stage 2 (KYC Integration Service)	D-KYC in progress at vendor	Yes
KYC_ROUTING_PENDING_MANUAL / ROUTING_PENDING_MANUAL	Section 6.6 (BRE fallback)	BRE unavailable at a routing checkpoint; awaiting manual routing	Yes
UNDERWRITING_PENDING	Stage 3 → Stage 4 handoff (Underwriting Service)	BRE eligibility check complete (Model A); case not yet picked up from queue	Yes
CREDIT_FILE_GENERATED	Stage 3 → Underwriting Service (Model B, Section 5.8)	BRE eligibility check complete; Credit File compiled and ready for Ops download	Yes
PENDING_EXTERNAL_DECISION	Underwriting Service (Model B, Section 5.8)	Ops has downloaded/shared the Credit File; awaiting the Finance Provider’s offline decision	Yes

Status	Set By (Stage/Service)	Meaning	Active ¹
UNDER_REVIEW	Stage 4 (Underwriting Service)	Case locked by an underwriter (Case Assignment, Stage 4)	Yes
DECISION_PENDING	Stage 4 (Underwriting Service)	Maker decision submitted, awaiting Checker	Yes
BRE_DECISION_PENDING_MANUAL	Section 6.6	BRE unavailable at credit decisioning; manual eligibility check required	Yes
CPV_UNREACHABLE	Stage 4	Customer unreachable for CPV; underwriter override note required to proceed	Yes
FINANCING_APPROVED	Stage 4 (Checker, Model A) or Section 5.3.4 (External Decision Capture, Model B)	Approved; proceeds to Offer Generation	No (terminal-forward)
REJECTED_BY_UNDERWRITER / REJECTED_FRAUD_FLAG	Stage 4	Rejected; application closed	No (terminal)
REWORK_REQUIRED	Stage 4	Routed back to specified stage (CPV/docs/underwriting)	Yes
PENDED_INFO_REQUESTED	Stage 4	Customer/Ops informed of outstanding requirement	Yes
OFFER_ACCEPTED	Stage 5, Part 1	Customer accepted the offer	Yes
OFFER_REJECTED / OFFER_EXPIRED	Stage 5, Part 1	Customer declined, or took no action before expiry	No (terminal)
CONTRACT_REWORK	Stage 5, Part 2	Ops rejected signed docs; customer re-uploads	Yes
PRE_DISBURSAL_COMPLETE	Stage 5, Part 2	All 4 pre-disbursal gates cleared	Yes
FINANCING_CREATED_LMS / FINANCING_DISBURSED	Stage 6	Booked / disbursed	No (terminal)
APPLICATION_WITHDRAWN / APPLICATION_EXPIRED	Cross-Cutting Exception Flows	Customer- or system-initiated exit	No (terminal)

¹ “Active” indicates the application is treated as an existing, in-progress record under Section 5.3.1 Scenario B (blocks a new application from being created for the same customer). “Terminal” statuses do not block a new application — Scenarios C/D apply instead.

Stage 1: Customer & Application Id Creation

Owning Microservice: Customer & Application Service (Section 5.3.1)

Actors: Customer, BLOX Portal, LOS Platform, Authentication Engine, SMS Gateway

Application level flow starts with Blox customer portal initiating application creation API.

Pre-Requisites

- LOS APIs are active and accessible.

Outcome

- Application submission request successfully accepted by M2P LOS

OTP Verification (Mobile Number Confirmation)

Before the application creation API is invoked, the customer's mobile number is confirmed via OTP through the BLOX Communication Gateway (Section 5.7.1; Section 13). This step is synchronous and gates access to application creation; it is executed by the Authentication Engine actor named above.

Scenario	Condition	System Behaviour
OTP Verified	Entered code matches within validity window	JWT issued; proceed to application creation
Incorrect OTP	Code mismatch, attempt < 5	OTP_INCORRECT; retry prompt shown; attempt counter incremented
Max Attempts Exceeded	attempt ≥ 5	OTP_LOCKED; channel locked 15 minutes; resend blocked for that window
OTP Expired	Entered after 5-minute validity window	OTP_EXPIRED; customer prompted to request a new code
Resend Limit Exceeded	> 3 resend requests within 15 minutes	Resend blocked; rate-limit message shown

Test Scenarios

Type	Scenario	Expected Output
Happy	Correct OTP entered on first attempt	OTP_VERIFIED; JWT issued
Negative	5 consecutive incorrect attempts	OTP_LOCKED; further attempts return 403 for 15 minutes
Negative	OTP entered after 5-minute validity window	OTP_EXPIRED; must request a new code
Edge	4th resend request within a 15-minute window	429 — resend rate limit exceeded

Field Validations (part of Application creation API)

Field	Mandatory	Validation
First Name	Yes	2-50 alphabetic characters
Last Name	Yes	2-50 alphabetic characters
Email	Yes	Valid email format and unique
Phone Number	Yes	Valid Qatar mobile number (+974, 8 digits)
QID	Yes	11 digits, valid checksum, age eligibility validation
Gender	Yes	Male, Female, Prefer Not to Say
Nationality	Yes	Valid nationality, auto-derived from QID where applicable

Test Scenarios

Scenario	Expected Result
Valid customer registration	Customer created successfully
Invalid QID	Validation error displayed
Duplicate email	Registration blocked
Invalid mobile number	Validation error displayed
Existing active customer	Duplicate customer message

Application Parameters — Vehicle Specifications (part of application creation API)

Field Name	Source	Mandatory/Non-Mandatory	Validations	List of values
Vehicle Price	Vehicle Master / API	Mandatory	Numeric, greater than 0. This is the "Asset Price" in Musharakah.	N/A
Vehicle Status	Vehicle Master / API	Mandatory	Must be 'New' or 'Used'. Affects financing rules.	New; Used
Make	Vehicle Master / API	Mandatory	Text. Must exist in the approved vehicle make list.	Vehicle Make Master
Model	Vehicle Master / API	Mandatory	Text. Must exist in the approved vehicle model list.	Vehicle Model Master
Year	Vehicle Master / API	Mandatory	Numeric, 4 digits. For used cars, must be within QCB's allowed age limit (e.g., max 5 years old).	N/A

Instalment & Application Parameters (part of application creation API)

Field Name	Source	Mandatory	Validations	List of values
Down Payment	Entered manually	Mandatory	Must meet QCB minimums (typically 20% of Vehicle Price for expatriates). This is the "Customer's Initial Share".	N/A
Finance Tenure (months)	Entered manually	Mandatory	Must be within QCB limits (e.g., max 72 months for expats, 48 for some visa types). Cannot exceed policy.	12; 24; 36; 48; 60; 72
Contract Amount	System Calculated	Mandatory	Vehicle Price - Down Payment. This is the "Bank's Share" of the asset.	N/A
Salary	Entered manually	Mandatory	Numeric. Must be verified via Salary Certificate.	N/A
Duration of Residence	Selected from dropdown	Mandatory	Critical for risk assessment, especially for expatriates.	Less than 6 months; 6-12 months; More than 12 months
Monthly Liabilities	Entered manually	Mandatory	Numeric. Used to calculate the Debt Burden Ratio (DBR), which must not exceed QCB limits (e.g., 50% for expats).	N/A

Field Name	Source	Mandatory	Validations	List of values
Employment Type	Selected from dropdown	Mandatory	Critical for credit policy and risk scoring.	Government or Semi-Government; Private Sector - Approved; Private Sector - Unlisted; Self-Employed

Plan & Schedule Summary (Sharia-Compliant Diminishing Musharakah) (part of application creation API)

Field Name	Source	Mandatory	Validations	List of values
Monthly Installment	System Calculated	Mandatory	Calculated based on Contract Amount, Tenure, and Profit Rate. Represents the monthly rent + unit purchase price to systematically buy out the Bank's share.	N/A
Total Periodic Payment Amount	System Calculated	Mandatory	Monthly Installment * Finance Tenure.	N/A
Blox Membership Fee	System Calculated	Non-Mandatory	Calculated based on selected membership plan.	N/A

Finance Provider Details (part of Application creation API)

Field Name	Source	Mandatory	Validations
Finance Provider Name	Finance Provider Master (Section 7.1, Finance Provider's Input & Policy Master)	Yes	Must exist in the approved Finance Provider Master; drives BRE Finance Provider-specific rules (Section 5.3.2)
Branch	Finance Provider Master	Yes	Must be an active branch under the selected Finance Provider
Finance Provider Unique ID	System-derived from Finance Provider Master	Yes	Read-only; not entered by customer or dealer

Note: the Engagement Mode and BRE Ownership configured against the selected Finance Provider (Section 5.8) determine which of the downstream flows in this document apply to the application. These attributes are read from the Finance Provider Master; they are not customer- or dealer-entered fields.

Dealer Details (part of Application creation API)

Field Name	Source	Mandatory	Validations
Dealer Name	Dealer Master	Yes	Must exist in the approved Dealer Master
Branch	Dealer Master	Yes	Must be an active branch under the selected Dealer
Sales Executive Name	Blox Portal session (Sales Executive role) / Dealer Master	Yes	Must match a provisioned Sales Executive under the selected Dealer branch
Executive Unique ID	System-derived	Yes	Read-only; used for dealer performance reporting (Section 7.1, Reporting & MIS Dashboard)

Vehicle Category (part of Application creation API)

Vehicle Status (New/Used, captured under Vehicle Specifications above) is independent of Vehicle Category below, which distinguishes cars from motorcycles.

Field Name	Mandatory	Validations	List of values
Vehicle Category	Yes	Must exist in the approved Vehicle Category list	Car; Motorcycle

Test Scenarios

Scenario	Expected Result
Eligible vehicle and financing parameters	Application created
Down payment below policy threshold	Validation error
Invalid tenure selection	Validation error
Vehicle age exceeds allowable limit	Validation error
Multiple submission attempts	Existing active application returned

Hard Stop: Application processing cannot proceed until all mandatory documents and consents have been successfully completed.

Required Documents (part of application creation API)

Document	Mandatory	Validation
Qatar ID (Front & Back)	Yes	Format, size, antivirus scan
Salary Certificate	Yes	Valid format and recency requirements
Bank Statements	Yes	Required period and valid format
Credit Bureau Report	Yes	Required period and valid format
Passport (Expatriates)	Conditional	Required for non-Qatari applicants
Additional Documents	No	Supported formats only

Document Controls

- Supported formats: PDF, JPG, PNG.
- Maximum file size: 5 MB per file.
- OCR processing initiated for identity documents.
- Document uploads stored with audit trails.

Mandatory Consents (part of application creation API)

Consent	Purpose
Credit Bureau Authorization	Bureau inquiry approval
Terms & Conditions Acceptance	Legal agreement acceptance
Digital KYC / Biometric Consent	Identity verification authorization
AML & Sanctions Screening Consent	Regulatory compliance screening

Consent Management Rules

- All consents are mandatory.
- Consent records are immutable after capture.
- Consent timestamp, customer identifier, device information, and session details are retained for audit purposes.
- Credit bureau integrations remain inaccessible until consent is provided.

Test Scenarios

Scenario	Expected Result
All documents uploaded successfully	Documents accepted
Unsupported file format	Upload rejected
File size exceeds limit	Upload rejected
Missing mandatory document	Processing blocked
Missing mandatory consent	Processing blocked
Valid uploads and consents	Application progresses to next stage

Once the application (vehicle, financing parameters, documents and consents) is submitted, the platform performs customer identification and de-duplication before the Customer ID and Application ID are finalized. The applicable outcome — new customer, existing active application, previously approved, or previously rejected — follows the Customer Identification & De-Duplication Scenarios (Section 5.3.1, Scenarios A–D), with negative-flow handling per Section 5.3.1’s companion Test Scenarios.

Final Outcome (response structure of Application creation API)

Entity	Status
Customer Profile	Created
Lead	Created
Application	Generated
Documents	Submitted
Consents	Captured
Audit Records	Logged
Next Stage	Identity Verification & Credit Assessment

Stage 2: DKYC Verification & Eligibility Assessment (AML, Sanctions, PEP Check)

Owning Microservice: KYC Integration Service (Section 5.3.3)

The customer already uploaded the documents and needs to complete the digital KYC checks.

1. D-KYC Execution — QID OCR & Biometric Liveness Matching

Integration Specification

Attribute	Specification
Endpoint	POST /api/initiate-dkyc (Portal → LOS middleware → KYC Vendor)
Sync/Async	Synchronous — customer waits on result
Timeout	15 seconds per attempt
Retry Policy	3 retries exponential backoff (2s, 4s, 8s); circuit breaker after 5 failures in 60s
Fallback on timeout	Status → DKYC_PENDING_MANUAL; route to Ops queue
Data Residency	All biometric payloads routed and stored in Qatar Cloud region ONLY
Face Match Threshold	≥ 80% (configurable in bre_parameters)

Decision Logic

Scenario	Condition	Status
OCR success + Liveness PASS	face_match_score ≥ 80% AND liveness = PASS	DKYC_VERIFIED → proceed to AML / PEP / Sanctions Screening
OCR fail — retryable	LOW_OCR_CONFIDENCE, attempt < 3	DKYC_RETRY — prompt with correction hint
OCR fail — exhausted	attempt ≥ 3	DKYC_PENDING_MANUAL → Ops queue
Spoof / Deepfake detected	failure_reason = SPOOF_DETECTED	DKYC_FRAUD_LOCKED — 24hr lock; MLRO P1 alert
Face mismatch	face_match_score < 60%	DKYC_FRAUD_LOCKED — escalate to Compliance

Test Scenarios

Type	Scenario	Expected Output
Happy	Clear QID, live selfie, face score 95%	DKYC_VERIFIED; proceed to AML / PEP / Sanctions Screening
Negative	Photo-of-a-photo spoof	SPOOF_DETECTED; channel locked 24hr; MLRO P1 alert
Negative	Face mismatch	FACE_MISMATCH; escalate Compliance
Negative	Expired QID	422; QID expiry < today; hard block
Edge	KYC service timeout	Retry x3; all fail → DKYC_PENDING_MANUAL; IT P3
Edge	Expatriate expired residency visa	Hard block — 'Residency permit expired'

2. AML / PEP / Sanctions Screening

HARD BLOCK ON CONFIRMED MATCH: A Confirmed AML match cannot be overridden by ANY user role including CRO/CEO. Profile lock enforced at DB AND API gateway level.

Attribute	Specification		
Endpoint	POST /api/aml/screen (LOS → AML Vendor)		
Sync/Async	Synchronous — blocks application progression		
Timeout	10 seconds		
Retry	3 retries, exponential backoff (1s, 2s, 4s)		
Watchlists (Indicative)	UN Sanctions, OFAC, EU, QCB local sanctions, Interpol, PEP (domestic + international)		
Fallback on failure	Status → AML_PENDING_REVIEW; Compliance queue; IT P2 alert		
Result	Confidence	LOS Action	Status
CLEAR	N/A	Proceed to Next Step	ELIGIBILITY_VERIFIED
POTENTIAL_MATCH	50-79%	Manual review — EDD required	AML_MANUAL_REVIEW
CONFIRMED_MATCH	≥ 80%	HARD BLOCK — auto-decline; profile locked; P1 MLRO alert	REJECTED_AML_FLAG

Note on Customer Communication — IMPORTANT: Customer must NOT be informed of the reason for AML decline. This is a legal requirement under QCB / FATF tipping-off rules. Show only generic 'Unable to proceed' message.

3. Residency Classification & Initial Eligibility Assessment

Eligibility Rules (ELIG001 Phase 1)

Rule	Qatari Threshold	Expat Threshold	On Fail
Age Check	18-65 years at contract end	21-60 years at contract end	Hard Reject (REJ_AGE)
Min Net Income	QAR 5,000/month (configurable)	QAR 7,000/month (configurable)	Soft Decline
Residency Status	N/A (Qatari national)	Must be ACTIVE; duration ≥ 6 months	Hard Reject
AML Status	Must be ELIGIBILITY_VERIFIED	Must be ELIGIBILITY_VERIFIED	Hard Block
D-KYC Status	Must be DKYC_VERIFIED	Must be DKYC_VERIFIED	Hard Block

Stage 3: Bank Statement Assessment, Salary Certificate OCR & BRE Decisioning (Backend)

Owning Microservice: Underwriting Service (inputs) invoking BRE Service (shared) for decisioning — Sections 5.3.2, 5.3.4, 6.2

The LOS platform runs automated checks and rule engine processing.

1. Bank Statement Ingestion & AI Parsing

Attribute	Specification
Endpoint	POST /api/bank-statement/analyze
Sync/Async	Asynchronous — LOS submits doc, polls webhook callback
Timeout	30s submission; processing up to 5 min via webhook
Webhook callback	POST /api/los/bank-statement/callback (vendor calls back)
Fallback (>10 min)	Status → BSA_MANUAL_REVIEW; route to underwriter for manual entry

Tamper Detection — Hard Escalation Triggers

- If PDF_METADATA_MISMATCH — creation date after declared statement period → status = BSA_TAMPER_SUSPECTED; immediate Fraud/MLRO P1 alert
- If FONT_SUBSTITUTION_DETECTED — different fonts on same document → status = BSA_TAMPER_SUSPECTED; immediate Fraud/MLRO P1 alert
- If PIXEL_MANIPULATION — rasterization artifacts on numbers → status = BSA_TAMPER_SUSPECTED; immediate Fraud/MLRO P1 alert
- If DIGITAL_SIGNATURE_INVALID — bank PDF signature tampered → status = BSA_TAMPER_SUSPECTED; immediate Fraud/MLRO P1 alert

2. Credit Bureau Report OCR

Owned by the Underwriting Service, consistent with Section 5.3.4. QCB Credit Bureau report is retrieved and OCR-parsed under the same operating model as Bank Statement Ingestion above.

Attribute	Specification
Endpoint	POST /api/credit-bureau/analyze (via QCB Credit Bureau integration, Section 2.2)
Sync/Async	Synchronous — 20 second timeout (Section 9.3)
Retry Policy	3 retries, exponential backoff; circuit breaker after 5 failures in 60s (Section 9.4)
Fallback on timeout/failure	Status → BUREAU_MANUAL_REVIEW; case routed to underwriter for manual bureau entry from a customer-submitted bureau report (Section 8, Stage 1 Required Documents)
No record found	FLAG THIN_FILE; BRE (CBS001, Section 9) proceeds using bank-statement-weighted evaluation rather than blocking the application

Salary Certificate & Income Verification (data point input by the ops user for bre calculation)

Extracted Field	Mandatory	Key Validation
Employer Name	Yes	Fuzzy match ≥ 80% against employer declared at Employer Master
Net Salary	Yes	Numeric > 0; primary figure used for DBR calculation

Extracted Field	Mandatory	Key Validation
Date of Certificate	Yes	≤ 90 days old; if older → STALE_CERT flag; manual review
Employer Stamp / Signature	Yes	OCR must detect stamp region; absence → MISSING_STAMP flag
Date of Joining	Yes	Must align with employment duration declared

Income Cross-Verification Rule

Variance = $|\text{salary_cert_net} - \text{bank_stmt_avg_credit}| / \text{salary_cert_net}$. If variance > 20% (configurable) → FLAG INCOME_MISMATCH. Use LOWER of the two figures for conservative DBR calculation. Underwriter must investigate in Case Assignment & Document Reconciliation step under Manual Review Stage.

3. BRE Execution — Credit Policy Decisioning

SHARIAH BOARD SIGN-OFF REQUIRED: SHARIA001 rule and fee routing logic require DIMINISHING MUSHARAKAH dual sign-off before any PROD change.

BRE Rule Pseudo-Code

ELIG001 — Eligibility & Knockout

```

IF Fraud_Sanctions_Match == TRUE:
    RETURN {status:"AUTO_DECLINED", code:"REJ_FRAUD", override:false}

IF Nationality == "QAT":
    IF Age < Config.Min_Age_Qatari OR Age > Config.Max_Age_Qatari:
        RETURN "REJ_AGE"
    IF NetIncome < Config.Min_Income_Qatari: RETURN "REJ_INCOME"
ELSE (Expatriate):
    IF ResidencyStatus != "ACTIVE": RETURN "REJ_RESIDENCY_INVALID"
    IF ResidencyDuration_months < Config.Min_Res_Expatriate: RETURN "REJ_RESIDENCY_DURATION"
    IF NetIncome < Config.Min_Income_Expatriate: RETURN "REJ_INCOME"
    IF EmployerCategory == "BLACKLISTED": RETURN "REJ_EMPLOYER_BLACKLIST" (non-overrideable)

```

CBS001 — QCB Credit Bureau

```

IF QCB_Report_Status == "NO_RECORD": FLAG("THIN_FILE"); RETURN "PROCEED_WITH_FLAG"
IF QCB_Score < Config.Score_Hard_Decline (550): RETURN "REJ_QCB_LOW"
IF QCB_Score < Config.Min_QCB_Score (650): RETURN "CONDITIONAL - guarantor required"
IF Default_History.active_defaults > 0: RETURN "REJ_ACTIVE_DEFAULT"
RiskTier = "A" if Score>=750, "B" if >=700, "C" if >=650

```

DBR001 — Debt Burden Ratio & Stress Test

```

Calculated_DBR = (ExistingObligations + ProposedEMI) / VerifiedNetIncome
Max_DBR = LookupDBRMatrix(Nationality, EmployerCategory, QCB_Score)
e.g. Expatriate Private = 0.50, Qatari Govt TierA = 0.70

IF Calculated_DBR > Max_DBR:
    IF excess <= 0.03: TRIGGER_EXCEPTION("DBR_BREACH_MINOR", tier:1) -- Head of Credit
    IF excess <= 0.05: TRIGGER_EXCEPTION("DBR_BREACH_MODERATE", tier:2) -- CRO
    IF excess > 0.05: TRIGGER_EXCEPTION("DBR_BREACH_SEVERE", tier:3) -- CRO+CEO
IF Calculated_DBR > Config.DBR_Hard_Cap (0.75): AUTO_DECLINE

IF FinancingAmount > Stress_HighTicket_Threshold:
    Stressed_Rental = ProposedEMI.Rental * (1 + 0.02) -- rate shock
    Stressed_Income = VerifiedNetIncome * (1 - 0.10) -- income shock
    Stressed_DBR = (Obligations + Principal + Stressed_Rental) / Stressed_Income
    IF Stressed_DBR > Config.Max_Stressed_DBR: TRIGGER_EXCEPTION tier:2

```

SELFEMP001 — Additional Credit Rules for Self-Employed Customers

This rule executes in addition to ELIG001, CBS001, DBR001 and EXC001 when the applicant's Employment Type is Self-Employed, since underwriting for self-employed applicants cannot rely solely on salary-based assessment. It assesses business sustainability, business cash-flow stability, income reliability, and repayment capacity.

```
IF EmploymentType == "SELF_EMPLOYED":
EXECUTE SELFEMP001

-- Mandatory validation checks: Active Commercial Registration (CR); Active Trade License; business
bank statements; personal bank statements where applicable; audited financial statements where
applicable; income verified through business activity and banking records
IF CommercialRegistrationStatus != "ACTIVE": RETURN "REJ_CR_INVALID"
IF TradelicenseStatus != "ACTIVE": RETURN "REJ_TRADE_LICENSE_INVALID"

IF BusinessOperatingPeriod_months < Config.Min_Business_Vintage (24):
FLAG("BUSINESS_VINTAGE_LOW"); ROUTE "REFER_CREDIT"

-- Business cash-flow assessment factors: cash-flow stability, revenue consistency, seasonal
fluctuations, related-party transfers, revenue trend deterioration, returned cheques, frequent
overdrafts, unexplained deposits
IF ConsecutiveNegativeCashFlowMonths >= 3: RETURN "REJ_NEGATIVE_CASHFLOW"
IF CashFlowStability == "LOW": FLAG("CASHFLOW_INSTABILITY"); ROUTE "REFER_CREDIT"

VerifiedIncome = MIN(DeclaredIncome, VerifiedBusinessIncome)
-- Only recurring, sustainable business income is considered. One-time gains, asset-sale proceeds and
other non-recurring income are excluded.

IF RevenueDecline_pct > 20: FLAG("DECLINING_BUSINESS"); ROUTE "REFER_CREDIT"

BusinessDBR = (ExistingObligations + ProposedInstallment) / VerifiedIncome
-- DBR evaluation and exception-tier routing on BusinessDBR continue to use DBR001 and EXC001 above;
not reimplemented here.

-- Mandatory referral conditions (ROUTE "REFER_CREDIT"): BUSINESS_VINTAGE_LOW, CASHFLOW_INSTABILITY,
DECLINING_BUSINESS, SOURCE_OF_FUNDS_REVIEW, HIGH_RELATED_PARTY_DEPENDENCY

-- Non-overrideable rejections (override:false always): REJ_CR_INVALID, REJ_TRADE_LICENSE_INVALID,
REJ_NEGATIVE_CASHFLOW, REJ_FRAUD, REJ_ACTIVE_DEFAULT, REJECTED_AML_FLAG

RETURN one of: "PROCEED" | "REFER_CREDIT" | "AUTO_DECLINED"
```

SHARIA001 — Diminishing Musharakah

```
EffectiveRentalRate = LookupRateCard(RiskTier, Tenure_Months, AssetType)
Blox_Share = TotalFinancingAmount

FOR month = 1 TO Tenure_Months:
    MonthlyRent = (Blox_Share * EffectiveRentalRate) / 12
    MonthlyPrincipal = TotalMonthlyInstallment - MonthlyRent
    ASSERT MonthlyRent > 0 AND MonthlyPrincipal > 0
    Blox_Share = Blox_Share - MonthlyPrincipal
    APPEND schedule_row(month, MonthlyRent, MonthlyPrincipal, Blox_Share)

-- Late fee: NEVER compound. Fixed QAR 150 → GL_LIAB_CHARITY (Gharamah — NOT income)
-- Early settlement: waive 100% unearned rent (Ibra) → GL_CONTRA_INC
```

EXC001 — Exception Routing

```
IF exception_count == 0: ROUTE "AUTO_REFER_CREDIT" (manual required per policy)
IF exception_count == 1 AND dbr_excess <= 0.03: ROUTE "HEAD_OF_CREDIT_RISK" (Tier 1)
IF exception_count == 2 OR dbr_excess <= 0.05: ROUTE "CRO" (Tier 2)
ELSE (count >= 3 or excess > 5%): ROUTE "CRO_AND_CEO" (Tier 3)
-- Maker cannot approve their own exception (DB constraint)
```

Test Scenarios

Type	Scenario	Expected Output
Happy	QCB 720, DBR 42%, expat private	RiskTier=B, 0 exceptions, REFER_CREDIT
Negative	Active QCB default	AUTO_DECLINED, REJ_ACTIVE_DEFAULT
Negative	DBR 78% (hard cap 75%)	AUTO_DECLINED, REJ_DBR_EXCEEDED
Negative	Employer blacklisted	AUTO_DECLINED, REJ_EMPLOYER_BLACKLIST (non-overrideable)
Edge	Thin file — expat new to Qatar	THIN_FILE flag; proceed heavy BSA/WPS weight
Edge	DBR 53% (expat cap 50%, 3% excess)	DBR_BREACH_MINOR; Tier 1 routing (not auto-declined)
Edge	SHARIA001 monthly principal negative (edge case)	REJ_SHARIA_STRUCTURE; product team notified

Engineering Checklist

- BRE implemented as independent microservice — not embedded in LOS monolith.
- All BRE parameters in bre_parameters table — configurable via SYS_ADMIN UI; no code deployment needed.
- SHARIA001 params require dual sign-off: Product Maker + DIMINISHING MUSHARAKAH Checker before PROD change.
- Simulation endpoint: POST /api/bre/simulate — dry-run against last 10,000 historical applications.
- Rollback: single-click via active_version_ptr in bre_config; auto-alert if rejection spike >10%.
- DB: bre_execution_log (WORM via Kafka → Elasticsearch); every execution logged with full input/output snapshot.

See Section 6 (BRE Architecture) for how this checklist is realized architecturally as a shared microservice.

Stage 4: Manual Review (Case Manager)

Owning Microservice: Underwriting Service (Section 5.3.4)

A Credit Risk Manager reviews all cases before approval or rejection.

1. Case Assignment & Document Reconciliation

- Underwriter fetches case via GET /api/cases/queue — sorted by exception tier DESC, wait time ASC, SLA breach risk.
- Optimistic lock applied: applications.locked_by = user_id, locked_at = now(). Second underwriter cannot edit until lock expires (15 min idle).
- Side-by-side dashboard: left = parsed/extracted data; right = original PDFs via fresh Blob SAS URL (15-min expiry, never cached).
- Underwriter reconciles extracted salary/obligations against PDF originals field-by-field.
- The queue is pre-filtered to the underwriter's assigned Finance Provider ID before the sort above is applied (Section 7.4); a BLOX-side underwriter with no Finance Provider restriction sees the unfiltered, all-Finance Provider queue.

2. Document Forensics & Tamper Detection

HARD STOP ON TAMPER: If any tamper flag detected: REJECTED_FRAUD_FLAG. Underwriter cannot override. MLRO P1 alert.

- POST /api/documents/verify-signatures
- Scans: PDF digital signature chain validity
- Scans: Creation/modification metadata vs stated statement period
- Scans: Font substitution fingerprint
- Scans: Pixel manipulation artifacts in image regions
- Scans: Copy-paste text layer anomalies

3. Contact Point Verification (CPV) — Discretionary Tele-Verification

Triggered at underwriter discretion based on employer type, DBR risk, or profile flags.

CPV Script Mandatory Checkpoints

#	Question	On Fail
1	Confirm full name and QID number	FLAG: identity mismatch
2	Are you aware you've applied for vehicle financing with BLOX?	FLAG: awareness failure — possible fraud/coercion
3	Do you wish to proceed with this financing?	Rejection — customer intent withdrawn
4	Confirm monthly salary and employer	FLAG: income/employer mismatch
5	Do you understand this is Shariah-compliant Diminishing Musharakah?	FLAG: Sharia disclosure gap

CPV NON-COMPLETION HANDLING: If the customer cannot be reached after 3 documented attempts across 2 business days across the contact channels on file, the case is flagged CPV_UNREACHABLE. The underwriter may either proceed on file-based evidence with a mandatory note, or PEND the case with a customer outreach request (SMS/Email, Section

13) to arrange a callback window. A CPV_UNREACHABLE case cannot reach FINANCING_APPROVED without an underwriter override note, which is logged to the audit trail.

4. Financial & Sharia Audit — DBR / FTV / Musharakah Validation

SHARIAH BOARD SIGN-OFF REQUIRED: Any change to DBR cap logic or Musharakah parameter validation requires DIMINISHING MUSHARAKAH sign-off.

Audit Item	Rule	On Breach
Final DBR ≤ QCB cap	Expat ≤50%; Qatari Govt ≤70%; Qatari Private ≤60%	Trigger exception; route to authority tier
Customer down payment confirmed	≥ 20% of asset value; matches payment receipt	Block — cannot proceed
Profit rate within rate card	Rate = LookupRateCard(RiskTier, Tenure, AssetType)	Rate card breach → CRO escalation
Rent calculated on BLOX's outstanding share only	Rent = Outstanding × AnnualRate / 12	SHARIA001 validation failure — auto-block
Late fee never compounds	Fixed QAR 150 → Charity GL only	DIMINISHING MUSHARAKAH escalation
Ibra calculated on early settlement	100% unearned rent waived → GL_CONTRA_INC	Pre-computed in schedule — mandatory
CAM Generation	Maker cannot be approved without CAM generation	Approval button disabled.

5. Maker-Checker Credit Decision

SEPARATION OF DUTIES: Maker cannot be Checker on the same case. Enforced at DB constraint, not UI only. Tier 3 requires CRO + CEO dual approval.

Checker Decision	New Status	Next Step
APPROVE	FINANCING_APPROVED	Offer Generation
REJECT	REJECTED_BY_UNDERWRITER	Notification to customer (QCB rejection code); closed
REWORK	REWORK_REQUIRED	Routed back to specified stage (CPV/docs/underwriting)
PEND	PENDED_INFO_REQUESTED	Customer/Ops informed of outstanding requirement
(Model B only) External decision	FINANCING_APPROVED or REJECTED_BY_UNDERWRITER	Same next step as Model A

captured — Section 5.3.4	(decision_source=EXTERNAL_FINANCE PROVIDER)	
Checker authority insufficient for financing amount (Section 1.5)	DECISION_PENDING (unchanged)	Auto-escalated to a Checker holding Head of Credit Risk authority; original Checker's action is not counted as the binding Checker decision
Type	Scenario	Expected Output
Happy	Clean case, Maker approves, Checker approves	FINANCING_APPROVED; offer generated
Negative	Maker tries to self-approve as Checker	403 — 'Same user cannot be Maker and Checker'
Negative	Tier 2 exception, Checker is not CRO	403 — 'Exception tier requires CRO approval'
Edge	Checker approves with blank remarks	422 — remarks minimum length enforced server-side

Approval-authority routing (Senior Manager vs. Head of Credit Risk, Section 1.5) is independent of, and in addition to, Tier 2/3 exception routing (Section 7.3, EXC001): authority routing is triggered by financing amount; exception routing is triggered by policy deviation. A single case may require both.

Stage 5: Offer Generation, Contracting & Agreement Review

Owning Microservice: Offer Management Service (offer generation & acceptance) and Legal Agreement Service (contract generation, signing, pre-disbursal setup) — Sections 5.3.5.1, 5.3.5.2

The customer receives and acts on the financing offer.

1. Offer Generation, Review & Customer Acceptance

HARD GATE — SHARIAH PRE-CONTRACT DISCLOSURE:

Offer screens MUST display full Musharakah breakdown (Asset Value, BLOX%/Customer%, Profit Rate, Rent per period, Total Financing Cost, Full Amortization Schedule) BEFORE acceptance button is enabled.

Field	Source	Mandatory Display
Total Asset Value (Vehicle Price)	Locked at Vehicle Selection Stage by Customer	Yes — 'Asset being acquired'
BLOX Financing Share (QAR + %)	Vehicle Price – Down Payment	Yes — BLOX's starting ownership
Customer Initial Contribution (QAR + %)	Down Payment confirmed	Yes — Customer's starting ownership
Annual Profit Rate	Rate card (RiskTier × Tenure)	Yes — expressed as % p.a. on outstanding share
Monthly Installment (Rent + Principal breakdown)	SHARIA001	Yes — component breakdown required
Total Financing Cost (sum of all rents)	SHARIA001 total	Yes — full transparency
Full Amortization Schedule	SHARIA001 output	Yes — downloadable PDF + portal table

Offer Expiry Date	System generated (7 days, configurable)	Yes
-------------------	---	-----

Test Scenarios

Type	Scenario	Expected Output
Happy	Customer reviews the full Musharakah breakdown and accepts within the validity window	OFFER_ACCEPTED; proceeds to Contract Generation
Negative	Customer explicitly rejects the offer	OFFER_REJECTED; application closed; customer may reapply per Section 5.3.1 Scenario D
Negative	Customer takes no action before the Offer Expiry Date (7 days, configurable)	OFFER_EXPIRED; application closed; customer must submit a new application to be re-offered
Negative	Customer attempts to accept after expiry (e.g., stale link/session)	409 — 'Offer expired'; acceptance blocked; must reapply
Edge	Customer requests a change to tenure or down payment after the offer is generated	Not supported as an in-place edit; treated as OFFER_REJECTED and a fresh application is required to regenerate offer parameters

2. Contract Generation, Signing, Verification & Pre-Disbursal Setup

Contract Bundle Required

Document	Mandatory	Signing Party
Diminishing Musharakah Agreement	Yes	Customer + BLOX/NBFC authorized signatory
Agency Agreement for Purchase (Wakala)	Yes	Customer
Key Facts Statement (KFS)	Yes	Customer (acknowledgment)
Credit Approval Memo (CAM)	Yes	NA
Wa'ad (Promise to Purchase)	Yes	Customer
Vehicle takaful Declaration	Yes	Customer

Pre-Disbursal Completion Gate

Atomic DB transaction — all 4 prerequisites checked simultaneously before status advances to PRE_DISBURSAL_COMPLETE: (1) All contracts VERIFIED, (2) Down payment SUCCESS, (3) Repayment mode REGISTERED, (4) takaful Opted/ACTIVE. Any single failure blocks status transition.

Type	Scenario	Expected Output
Happy	All docs signed, payment success, mandate registered	PRE_DISBURSAL_COMPLETE
Negative	Down payment card declined	Payment FAILED; retry allowed; status stays OFFER_ACCEPTED
Negative	Ops rejects signed docs (missing page)	CONTRACT_REWORK; customer re-uploads

Type	Scenario	Expected Output
Negative	takaful coverage < vehicle value	422 — 'Sum insured must be ≥ vehicle value'

Stage 6: Final Ops Review, LPO Generation & Contract Creation

Owning Microservice: Disbursement Service (Section 5.3.7)

The final administrative steps before vehicle handover.

1. Final Ops Review, LMS Booking, LPO Dispatch & Vehicle Handover

HARD GATE: LPO cannot be generated without: VIN confirmed + all contracts verified + down payment receipted + repayment mode registered + takaful opted/active.

Ops Final Checklist

Checklist Item	Verification Method	On Fail
All Sharia contracts signed & verified	contracts.all_verified = true	PENDED_OPS_REVIEW
Down payment receipted	payments.status = SUCCESS	Block
Repayment mode registered	repayment_mandates.status = REGISTERED	Block
Vehicle VIN / Chassis / Engine confirmed	FIN_OPS manual verification vs dealer quotation	Return to dealer
takaful opted/active, coverage ≥ vehicle value	takaful_policies.status = ACTIVE	Block

LMS Integration

Synchronous call to LMS: POST {LMS_BASE_URL}/api/contracts/create. Timeout: 30 seconds. Idempotency key: 'LMS-{ApplicationID}'. On success: receive LMS_Account_ID; status → FINANCING_CREATED_LMS. On HTTP 500 or timeout: DO NOT assume success. Rollback to PRE_DISBURSAL_COMPLETE. IT P2 alert. Manual retry queue. Customer NOT notified until resolution.

Note: Add a qualifying note: LMS integration is currently modeled as a single POST call. If LMS requires multi-step booking, the payload will be decomposed accordingly. Final decomposition to be confirmed with the LMS team before FSD sign-off.

LPO Dispatch

Asynchronous after LMS success. POST to BLOX Middleware with LPO PDF (cryptographically signed with BLOX/NBFC private key via Azure Key Vault HSM). On acknowledgment: status → FINANCING_DISBURSED. Customer notified 'Vehicle ready for pickup'. Dealer notified via email with LPO PDF. On failure: 3 retries exponential backoff; fallback = manual email LPO dispatch by FIN_OPS.

Type	Scenario	Expected Output
Happy	All checklist pass; LMS 201; LPO acknowledged	FINANCING_DISBURSED; all parties notified
Negative	VIN mismatch	PENDED_OPS_REVIEW; dealer must reconfirm

Type	Scenario	Expected Output
Negative	LMS HTTP 500	Rollback to PRE_DISBURSAL_COMPLETE; IT P2; manual retry
Negative	LMS timeout (30s)	DO NOT assume success — rollback; check idempotency key
Edge	LPO dispatch fails — all retries exhausted	FIN_OPS manual email LPO to dealer

Cross-Cutting Exception Flows (Applicable Across Stages 1–6)

The flows below are not tied to a single stage; they can occur while an application is in any of Stages 1–6 and are handled the same way regardless of where the application currently sits.

A. Customer-Initiated Withdrawal / Cancellation

- The customer (or the Dealer acting on the customer’s behalf) may request withdrawal at any point before FINANCING_DISBURSED, via the Blox Portal or by contacting BLOX Ops.
- System behaviour: status → APPLICATION_WITHDRAWN; all pending BRE, underwriting, and offer actions are cancelled. Any down payment collected but not yet applied is refunded through BLOX’s refund process, which sits outside the LOS system boundary (Section 1.3).
- Withdrawal after contract signing but before disbursement carries the same status, and additionally requires OPS_ADMIN or FIN_OPS confirmation that no funds have moved and that any signed documents are voided; the action is logged to the audit trail.
- Withdrawal is only ever set on an explicit customer or authorized Ops action — it is never inferred from inactivity (see Part B, Application Auto-Expiry, for the inactivity-driven flow).

Type	Scenario	Expected Output
Happy	Customer withdraws while at KYC Pending	APPLICATION_WITHDRAWN; no further processing
Negative	Withdrawal requested after FINANCING_DISBURSED	Blocked — ‘Cannot withdraw a disbursed contract; refer to LMS servicing’
Edge	Withdrawal requested after contract signed, before down payment receipted	APPLICATION_WITHDRAWN; contract voided; Ops notified to reconcile any partial payment

B. Application Auto-Expiry (Inactivity / Stale Applications)

- If an application remains in a pre-approval status (Created, Submitted, KYC Pending/In Progress) with no customer or system activity for 30 days (configurable BRE parameter), the platform sets status → APPLICATION_EXPIRED and notifies the customer.
- Applications in Underwriting Pending / Under Review / Decision Pending statuses are excluded from auto-expiry — these are Ops-owned queue states, not customer-inactivity driven; SLA breach alerting (Section 10) applies instead.
- An expired application cannot be resumed; the customer must submit a new application, which follows Section 5.3.1 Scenario D (previous application closed → reuse Customer ID, new Application ID).

Type	Scenario	Expected Output
Happy	No activity for 30 days at Submitted status	APPLICATION_EXPIRED; SMS/Email notice sent
Negative	Customer resumes on day 31 via a saved portal link	410 — 'Application expired'; must start a new application
Edge	Application sits at Underwriting Pending for 45 days	NOT auto-expired; SLA breach alert to Case Manager instead (Section 10)

9. BRE Governance — Fee Master & Parameter Dictionary

Content preserved unchanged from FSD v2.0. This is the operational governance detail referenced architecturally in Section 6.

9.1 Sharia-Compliant Fee & Charge Master

CRITICAL SHARIA RULE: Administrative Charges (LATE_01) MUST route to GL_LIAB_CHARITY — donated to approved charitable causes. Any system change routing Administrative Charges to income GL requires immediate DIMINISHING MUSHARAKAH escalation.					
Fee Code	Category	Calculation	Indicative Fee Cap (Configurable)	Shariah Treatment	GL Code
PROC_01	Processing	Lesser of 1% finance amount or QAR 1,000 fixed	QAR 1,000	Cost_Recovery_Income — permissible as admin	GL_INC_FEE
LATE_01	Late Payment	Fixed QAR 150/month — NEVER per-diem or compounding	QAR 150/month	Charity_Gharamah — MUST be charitable donation	GL_LIAB_CHARITY
EARLY_01	Early Settlement Admin	Fixed QAR 200 admin cost recovery	QAR 200	Cost_Recovery_Income	GL_INC_FEE
IBRA_01	Ibra — Early Settlement Rebate	100% unearned future rent waived — mandatory Sharia	N/A	Ibra_Rebate → customer credit	GL_CONTRA_INC

9.2 Configurable BRE Parameters (Never Hardcode)

Parameter Key	Default	Description	Change Authority
Min_QCB_Score	650	Minimum QCB score for standard approval	CRO sign-off
Score_Hard_Decline	550	Score below which auto-decline regardless	CRO sign-off
Max_DBR_Expat_Private	0.50	Max DBR for expat in private sector	CRO + QCB alignment
Max_DBR_Qatari_Govt_TierA	0.70	Max DBR for Qatari govt employee (QCB≥700)	CRO + QCB alignment
DBR_Hard_Cap	0.75	Absolute ceiling — auto-decline above this	CRO + CEO + QCB
Stress_Rental_Increase	0.02	+2% rate shock in stress test	CRO sign-off

Parameter Key	Default	Description	Change Authority
Stress_Income_Reduction	0.10	10% income reduction in stress test	CRO sign-off
Min_Cust_Contribution_New	0.20	Min 20% down payment for new vehicles	CRO + QCB
Max_Tenure_Expat	48	Max tenure months for expatriates	CRO + QCB
Max_VehicleAge_End	10	Vehicle max age at tenure end	CRO sign-off
Max_FTV_Expat	0.80	Max Finance-to-value for expatriates	CRO + QCB
Income_Variance_Threshold	0.20	Max variance salary cert vs bank statement	CRO sign-off
Offer_Expiry_Days	7	Days before offer expires	Product Manager

9.3 Integration Specifications — Integration Matrix

Service	Sync/Async	Timeout	Retry	Fallback
QID OCR / D-KYC	Sync	15s	3× exp backoff	Manual ops queue
AML/PEP (Zigram)	Sync	10s	3× exp backoff	Compliance review queue
Bank Statement AI	Async (webhook)	Sub:30s / Processing:5min	3× on submission	Manual underwriter entry
Salary Cert OCR	Sync	15s	3×	Manual entry
QCB Credit Bureau	Sync	20s	3×	Manual review; thin-file flag
WPS (via BLOX MW)	Sync	15s	3×	Rely on BSA + salary cert
Payment Gateway	Async (webhook)	Sub:10s	3×	PDC fallback
BLOX Comms (SMS/Email)	Async	10s	3×	Queue; IT P3 if all fail
LMS	Sync	30s	Manual retry	Rollback to PRE_DISBURSAL_COMPLETE
LPO / Dealer (BLOX MW)	Async	30s	3×	Manual email LPO dispatch

9.4 Universal Integration Rules

- Idempotency Keys: All external calls include Idempotency-Key header = '{Service} - {ApplicationID - AttemptN}'. Prevents duplicate Tawidh/records on retry.
- Circuit Breaker: Open after 5 consecutive failures in 60s; half-open probe after 30s. Apply to all external calls.
- Data Residency: All API traffic to KYC, AML, BSA vendors must route through Qatar cloud region. Contractual requirement.
- TLS: All outbound calls use TLS 1.2 minimum; TLS 1.3 preferred. Certificate pinning for KYC and AML endpoints.
- Webhook Security: Incoming webhooks (BSA etc) verified via HMAC-SHA256 signature before processing.
- Secrets: All API keys in Cloud (e.g. Azure Key Vault) — injected at runtime. Never in code or config files.
- Logging: Every call logged (request_id, endpoint, payload hash, response status, latency) WORM via Kafka.

These integration rules apply uniformly to service-to-vendor calls and to the service-to-service integration points defined in Section 5.6.

10. Performance, Scalability & SLA

10.1 Business SLA

Approval TAT: < 24 hours for standard salaried STP-eligible applications (lead submission → LPO generation).

10.2 Technical Performance SLAs

Metric	Target	Condition
API Response Time (Baseline)	< 2.0s P95	100 concurrent users
API Response Time (Peak)	< 3.0s P95	200 concurrent users
DB Query (Credit Assessment)	< 500ms avg	1,000 concurrent queries
BRE Execution (excl. external APIs)	< 800ms P95	Standard application
Auto-Scaling Response	< 5 minutes	Spike from baseline to 500 users
Error Rate at Peak	< 0.1%	500 concurrent users
Platform Uptime	99.9%	Annualized production
RTO (Disaster Recovery)	4 hours	Primary region failure
RPO (Data Loss Tolerance)	1 hour	Any failure scenario

10.3 Annual Origination Projections

Year	Disbursed Contracts
Year 1	~1,000
Year 2	~1,300
Year 3	~1,820
Year 5	~4,100

11. Information Security & Compliance

11.1 Field-Level Encryption (FLE)

Algorithm: AES-256-GCM. Key management via Cloud (e.g. Azure Key Vault) (HSM). Encryption applied at application layer before DB write. Plaintext data never reaches DB.

Entity	Field	Storage	UI Display
Customer	QID Number	AES-256-GCM + SHA-256 blind index	Masked: XXXXXXX1234
Customer	Passport Number	AES-256-GCM + SHA-256 blind index	Masked: XXXXX678

Entity	Field	Storage	UI Display
Customer	Full Name	AES-256-GCM encrypted	Shown to authorized roles only
Bank Info	Customer IBAN	AES-256-GCM encrypted	Masked: QAXXXXXXXXXXXXXXXXXX1234
Customer	Phone Number	AES-256-GCM encrypted	Masked: +974 XXXX X456

11.2 Session Management

Parameter	Value	On Breach
Idle Timeout	10 minutes	Auto-terminate; clear tokens; redirect to login
Idle Warning	2 min before timeout	Countdown modal
Absolute Timeout	8 hours	Force-terminate regardless of activity
Max Concurrent Sessions	1 per user	Second login attempt blocked
JWT Algorithm	HS256 (RS256 for service-to-service)	Invalid/expired → 401

11.3 Step-Up Authentication

High-Risk Operation	Step-Up Method
Authorize LPO Generation	OTP to registered mobile
Modify IBAN / Bank Account	OTP + manager approval
Manual Credit Override Tier 2/3	OTP to authorized approver's mobile
Fee Waivers > QAR 500	OTP + dual approval
Unmask PII (Compliance Officer only)	OTP; action logged immutably
Generate/Download Credit File (Section 5.8)	OTP to OPS_ADMIN's registered mobile; action logged immutably with recipient Finance Provider name and timestamp

11.4 PDPPL Compliance Matrix (Qatar Law No. 13 of 2016)

System Feature	PDPPL Article	Justification
AES-256-GCM FLE on QID/IBAN	Art. 13 — Precautions to Protect Data	Protects PII even in DB breach
RBAC + Dynamic PII Masking	Art. 14 — Admin & Technical Measures	Need-to-know at rendering layer
Audit trail for PII decryption	Art. 14 §3 — Tracking	Monitor unauthorized access attempts
TLS 1.2+ for all API transit	Art. 11 — Lawful Processing	Secures PII in transit to bureau/LMS
Data stored in Qatar Cloud only (e.g. Azure)	Art. 16 — Cross-Border Transfer	No PII leaves Qatar
10-year WORM retention	Art. 18 — Retention Period	QCB regulatory mandate

12. Infrastructure & Deployment

Component	Cloud Service	Specification
Compute	Kubernetes (e.g. Azure Kubernetes Service)	Microservices; each domain independently scalable via HPA
Relational DB	Database (e.g. Azure SQL Database)	Transactional data; audit trails; encrypted configs; zone-redundant
Object Storage	File Storage (e.g. Azure Blob Storage)	Customer documents; Qatar region only; AES-256 at rest
Cache	Data Cache (e.g. Azure Redis Cache)	Session tokens; OTP rate limiting; BRE parameter cache
Message Queue	Kafka (e.g. Azure Service Bus / Kafka)	Async events; audit logs; BSA webhooks; decouples microservices
Network Security	WAF (e.g. Azure WAF + Load Balancer)	Public: WAF-protected customer APIs. Internal: VPN/IP whitelist
Secrets / Keys	Key Mngt (e.g. Azure Key Vault) (HSM)	AES keys; service credentials; LPO signing keys
Observability	Monitoring (e.g. Azure Monitor + Datadog/ELK)	APM; structured logging; alerting; K8s pod health

12.1 Shared Responsibility Matrix

Area	Owner	Scope
SaaS Application Logic	M2P	LOS software, BRE, bug fixes, feature config
Cloud Infrastructure	BLOX	Cloud provisioning, networking, storage, compute (e.g. Azure)
Infrastructure Security	BLOX	Firewalls, WAF rules, OS patching, physical security
CI/CD & Deployments	BLOX DevOps	Pipeline management; deployment scheduling; rollback execution

13. Communications & Notifications

Channel	Use Case	Fallback
SMS	OTP; payment reminders; overdue escalations	Email fallback for OTP
Email	Dense legal/financial content: approval letters, contracts, schedules	Portal in-app notification
In-App (Portal)	Real-time status updates	N/A — always delivered

13.1 Key Notification Templates

Trigger	Channel	Template
OTP request	SMS (Email fallback)	Your BLOX verification code is {{OTP}}. Expires in 5 minutes. Do not share.
Application submitted	SMS	Hi {{FirstName}}, your application is submitted and under review.
Application approved	SMS + Email	Hi {{FirstName}}, your financing is approved. Review your offer in the app.
Application rejected	SMS	Hi {{FirstName}}, we are unable to proceed at this time. (No reason disclosed — QCB tipping-off rule)
LPO released	SMS (Customer) + Email (Dealer)	Congratulations {{FirstName}}! Your vehicle is ready for pickup.
Payment overdue 1-3 DPD	SMS	BLOX Notice: QAR {{Amount}} overdue. Please make payment.

14. Testing, QA & Acceptance Criteria

Test Stream	Scope	Tool / Method
Unit & Functional	All API endpoints; positive + negative + boundary	Jest / JUnit; automated CI (Continuous Integration) on every PR (Pull Request)
Integration (API)	D-KYC, AML, BSA, QCB Credit Bureau, LMS, PG — end-to-end payload validation	Postman / Karate; mocked vendor sandboxes
BRE Validation	All rule combos: DBR edge cases, score bands, employer categories	Dedicated BRE test harness with 200+ parametrized scenarios
Performance & Load	Baseline/Peak/Stress; K8s auto-scaling validation	k6 / JMeter; Load Testing (Azure)
Security (VAPT)	OWASP Top 10; JWT expiry; FLE (Field-Level Encryption); RBAC bypass; IDOR (Insecure Direct Object Reference)	Annual external pentest + automated SAST/DAST (Static/Dynamic Application Security Testing) in CI
UAT & Sharia Sandbox	Client-led; Musharakah schedule accuracy; QCB sandbox; DIMINISHING MUSHARAKAH	Manual with synthetic data only — no real PII

14.1 Production Hard Gates (Go / No-Go)

Gate	Criteria
Zero Critical/High Defects	No open P1/P2 defects in issue tracker
Performance SLA	P95 <2-3s @ 100 users; <3-4s @ 200 users
Consent Hard Block	D-KYC and bureau APIs blocked without explicit consent — tested via direct API injection
Maker-Checker Enforcement	Same user cannot approve own submission — tested with API injection attempt
Shariah Pre-Contract Disclosure	Full Musharakah breakdown displayed before acceptance button enabled
LPO Pre-Condition Check	LPO blocked if VIN/takaful/contracts/down payment incomplete
AML Hard Block	Confirmed AML match cannot be overridden — tested with DB state manipulation attempt
DIMINISHING MUSHARAKAH Sign-Off	Written DIMINISHING MUSHARAKAH approval for SHARIA001 and all contract templates
Audit Integrity	Every material state change logged WORM — verified by injection test
Data Residency	Network trace confirms zero data egress outside Qatar Cloud region (e.g. Azure)

15. Reconciliation & BRE Assurance

The Recon Module runs asynchronously (every 15 minutes via event queue) after each BRE execution. It re-computes the same inputs using an independent rule script and compares output to the primary BRE decision. Any mismatch halts application progression and routes to Model Risk Review Queue.

Architecturally, this module operates against the shared BRE Service (Section 6) and is independent of any single workflow-owning service — it is what allows BRE's decisions to be trusted as a shared dependency across Customer & Application, KYC Integration, and Underwriting Services.

15.1 Hard Reject Assertions

Scenario	Expected BRE Status	Mismatch Action
Document tampering	REJECTED_FRAUD_FLAG	P1 alert; application locked
QCB score below hard decline	AUTO_DECLINED REJ_QCB_LOW	P1 alert if application advanced
Confirmed AML/Sanctions match	REJECTED_AML_FLAG	P1 MLRO alert; LPO blocked
D-KYC liveness failure	DKYC_FRAUD_LOCKED	P1 alert; profile lock verified
DBR above absolute cap	AUTO_DECLINED REJ_DBR_EXCEEDED	P2 alert; Credit Ops investigation
Sharia structure breach	REJ_SHARIA_STRUCTURE	P1 DIMINISHING MUSHARAKAH alert

15.2 Daily Recon Report (Automated EOD)

- Total BRE Executions in period
- Recon Mismatch Rate (alert if >0.1%)
- Hard Rejects by Reason Code (REJ_QCB_LOW, REJ_DBR_EXCEEDED, REJ_AML, etc.)
- Recon Exceptions Resolved vs Open in Model Risk queue

16. Appendix

16.1 Glossary

Term	Definition
BRE	Business Rule Engine — core decisioning microservice evaluating credit policies, DBR, and Sharia rules
DBR	Debt Burden Ratio = (Existing obligations + Proposed EMI) / Verified net monthly income. QCB-capped.
Diminishing Musharakah	BLOX and customer jointly own asset; customer buys BLOX's share monthly via equity installments + rent on outstanding share
Hard Reject	Auto-declined by BRE on critical policy/regulatory/fraud basis. Non-overridable.
Ibra	Mandatory Sharia rebate — 100% unearned future rent waived on early settlement
DIMINISHING MUSHARAKAH	Internal Shariah Supervisory Committee — signs off any Musharakah logic or fee routing change
LMS	Loan Management System — downstream core banking post-origination
LPO	Local Purchase Order — issued to dealer authorizing vehicle release
FTV	Finance-to-Value = Financing Amount / Vehicle Price
MLRO	Money Laundering Reporting Officer — investigates AML/PEP/Sanctions hits
PDPPL	Qatar Personal Data Privacy Protection Law No. 13 of 2016
QCB	Qatar Central Bank — primary financial regulator
RPO	Recovery Point Objective — max tolerable data loss. Target: 1 hour.
RTO	Recovery Time Objective — max tolerable downtime. Target: 4 hours.
WORM	Write Once Read Many — audit log immutability. No UPDATE/DELETE on audit tables.
Thin File	No QCB credit history — typically expats new to Qatar. BRE bypasses score rule.
UN / OFAC / EU / Interpol	International sanctions/watchlist authorities screened during AML/PEP checks
PEP	Politically Exposed Person: Domestic PEP, International PEP
FATF	Financial Action Task Force
Microservice	An independently deployable LOS service owning a bounded business capability
Shared Microservice	A service, such as BRE, invoked by multiple owning services rather than sequenced as a workflow stage,

16.2 API Catalog

API Endpoint	Method	Purpose	Auth	Owning Microservice
/api/create-customer	POST	Create customer profile (CustomerID)	Portal service token	Customer & Application Service
/api/auth/send-otp	POST	Dispatch OTP	Public (rate-limited)	Customer & Application Service
/api/auth/verify-otp	POST	Verify OTP; issue JWT	Public	Customer & Application Service
/api/leads	POST	Create lead + deduplication check	Public (rate-limited)	Customer & Application Service
/api/applications	POST	Create application with vehicle + financing params	JWT (CUST)	Customer & Application Service
/api/documents/upload	POST	Upload KYC/financial documents	JWT (CUST)	Customer & Application Service
/api/dkyc/initiate	POST	Trigger D-KYC biometric verification	JWT (CUST)	KYC Integration Service
/api/compliance/aml-screen	POST	AML/PEP/Sanctions screening	Internal service	KYC Integration Service
/api/bsa/submit	POST	Submit bank statement for AI parsing	Internal service	Underwriting Service
/api/los/bsa/callback	POST	Receive BSA webhook from vendor	HMAC webhook	Underwriting Service
/api/salary-cert/analyze	POST	OCR salary certificate	Internal service	Underwriting Service
/api/bre/execute	POST	Run BRE decisioning	Internal service	BRE Service (shared)
/api/bre/simulate	POST	Dry-run BRE change against historical data	SYS_ADMIN JWT	BRE Service (shared)
/api/cases/queue	GET	Fetch next prioritized case for underwriter	JWT (CRCH_UW)	Underwriting Service
/api/cases/{id}/maker-recommendation	POST	Submit Maker decision and notes	JWT (CRCH_UW)	Underwriting Service
/api/cases/{id}/checker-decision	POST	Submit binding Checker decision	JWT (CRCH_CHK/BLOX_RISK)	Underwriting Service
/api/credit-file/generate	POST	Compile Credit File (application data + customer documents + KYC + bank/bureau analysis + BRE	Internal, Ops-triggered	Underwriting Service

API Endpoint	Method	Purpose	Auth	Owning Microservice
		output) for a Model B Finance Provider (Section 5.8)		
/api/credit-file/{id}/download	GET	BLOX Ops downloads the compiled Credit File for handoff to the Finance Provider (Section 5.8)	JWT — OPS_ADMIN/FIN_OPS only	Underwriting Service
/api/offers	POST	Generate financing offer post-approval	Internal service	Offer Management Service
/api/offers/{id}/response	POST	Customer accepts/rejects offer	JWT (CUST)	Offer Management Service
/api/contracts/generate	POST	Generate Sharia contract PDF bundle	Internal service	Legal Agreement Service
/api/contracts/{id}/upload-signed	POST	Upload wet-signed contract scans	JWT (CUST)	Legal Agreement Service
/api/repayment/setup	POST	Register auto-debit mandate or PDC	JWT (CUST)	Legal Agreement Service
/api/payments/down-payment	POST	Initiate down payment via PG (Integration Owner — BLOX)	JWT (CUST)	Legal Agreement Service
/api/ops/authorize-lpo	POST	Ops final checklist + LPO authorization	JWT (FIN_OPS)	Disbursement Service
/api/lms/create-contract	POST	Synchronous LMS contract booking	Service-to-service (mTLS)	Disbursement Service
/api/lpo/dispatch	POST	Dispatch LPO to dealer via BLOX middleware	Service-to-service	Disbursement Service
/api/applications/{id}/status	GET	Portal queries LOS for current status	JWT (CUST/Portal)	Customer & Application Service

16.3 Service-to-API Traceability Matrix

A summarized count view of Section 16.2, provided to support engineering decomposition planning.

Microservice	API Count	Representative Endpoints
Customer & Application Service	7	/api/create-customer, /api/leads, /api/applications, /api/documents/upload, /api/applications/{id}/status
BRE Service (shared)	2	/api/bre/execute, /api/bre/simulate
KYC Integration Service	2	/api/dkyc/initiate, /api/compliance/aml-screen
Underwriting Service	8	/api/bsa/submit, /api/los/bsa/callback, /api/salary-cert/analyze, /api/cases/queue, /api/cases/{id}/maker-recommendation,

Microservice	API Count	Representative Endpoints
		/api/cases/{id}/checker-decision, /api/credit-file/generate, /api/credit-file/{id}/download
Offer Management Service	2	/api/offers, /api/offers/{id}/response
Legal Agreement Service	4	/api/contracts/generate, /api/contracts/{id}/upload-signed, /api/repayment/setup, /api/payments/down-payment
Disbursement Service	3	/api/ops/authorize-lpo, /api/lms/create-contract, /api/lpo/dispatch

List of Pending Items

S.No	Name / Item	Applicability / Purpose	Owner	Integration Type / Category	Status
1	Qatar ID (Qatari Nationals) / Residency Permit (Expatriates)	All Applicants	BLOX	Standard Document	Pending
2	Passport Copy	All Applicants	BLOX	Standard Document	Pending
3	Valid Driving License	All Applicants	BLOX	Standard Document	Pending
4	Proof of Residence (Utility Bill / Rental Agreement)	All Applicants	BLOX	Standard Document	Pending
5	Credit Bureau Report	All Applicants	BLOX	Standard Document	Pending
6	Vehicle Quotation / Evaluation Report	All Applicants	BLOX	Standard Document	Pending
7	Signed Consent Form (Template for (Credit Check & Data Verification, KYC initiation and storage)	All Applicants	BLOX	Standard Document	Pending
8	Signed Diminishing Musharakah Agreement (Template:- Financer's specific) KFS & CAM template	All Applicants	BLOX	Standard Document	Pending
9	Salary Certificate (Not Older Than 30 Days)	Employed Applicants	BLOX	Additional Document	Pending
10	6 Months Bank Statements (Salary Account)	Employed Applicants	BLOX	Additional Document	Pending
11	Employment Contract	Expatriate Employed Applicants	BLOX	Additional Document	Pending
12	Valid Commercial Registration (CR)	Self-Employed Applicants	BLOX	Additional Document	Pending
13	12 Months Personal Bank Statements	Self-Employed Applicants	BLOX	Additional Document	Pending
14	12 Months Business Bank Statements	Self-Employed Applicants	BLOX	Additional Document	Pending

15	Trade License	Self-Employed Applicants	BLOX	Additional Document	Pending
16	Audited Financials (If Partnership Contribution > QAR 200,000)	Self-Employed Applicants	BLOX	Additional Document	Pending
17	Tax Card or Equivalent	Self-Employed Applicants	BLOX	Additional Document	Pending
18	Business Asset Valuation	Self-Employed Applicants	BLOX	Additional Document	Pending
19	Credit Report from Home Country	New Residents without QCB History	BLOX	Conditional Document	Pending
20	6 Months Continuous Salary History in Qatar	New Residents without QCB History	BLOX	Conditional Document	Pending
21	Additional Security Deposit (10% of Financing Amount)	New Residents without QCB History	BLOX	Conditional Requirement	Pending
22	Customer& Application Creation API	Customer application submission	M2P	Application submission API	Pending
23	Digital KYC / Identity Verification Provider	Identity verification and biometric validation	M2P	API	Pending
24	Credit Bureau Report Review Capability	PDF viewer and manual review capability	M2P	Internal Feature/ Qatar Credit Bureau API Integration Doc	Pending
25	Bank Statement Analysis Service	Income and transaction assessment	M2P	Internal Feature/API	Pending
26	AML / Compliance Screening Provider	AML, PEP and Sanctions screening	M2P	API	Pending
27	Credit Bureau Integration	Bureau integration	BLOX	Qatar Credit Bureau API Integration Doc	
28	WPS (Wage Protection System)	Salary verification	BLOX	Document sample	Pending
29	Cloud Service Provider	LOS hosting and infrastructure	M2P / BLOX	Infrastructure Platform	Pending
30	Shariah Advisory Board	Shariah compliance governance	BLOX	Governance	Pending
32	SMS Gateway	OTPs and customer notifications	M2P / BLOX	API	Pending
33	Email Service Provider	Email notifications and document delivery	M2P / BLOX	API	Pending
34	Financer Master List	All configured Finance Providers / NBFCs	BLOX	Master Data	Pending

35	Dealer Master List	All approved dealers and branches	BLOX	Master Data	Pending
36	Complete API Specification Document	All APIs used within LOS ecosystem	M2P & BLOX	API Documentation	Pending
37	Finance Provider Product Configurations	List of finance providers, Workflow to be followed, BRE to be run for underwriting	BLOX	Master Configuration	Pending
38	Dealer Branch & Sales Executive Master	For authorising application submission API while onboarding journey. When Dealer is Financer as well.	BLOX	Master Configuration	Pending

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